

Particle Physics

Based on lectures by Thomas Rytov

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These notes are not endorsed by the lecturers. All errors are mine.

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1. Introduction

As physicists we want to figure out what matter is made of. This is what particle physics attempts to do. We will find that the fundamental building blocks of matter can be classified into a relatively small number of types.

This course tries to get a qualitative understanding of the Standard Model, which is our current best theory for understanding the nature of particles and their interactions. We will not do any explicit quantum field theory, but we will cover Feynman diagrams and their associated Feynman rules.

2. The Zoo

2.1. The “classical” particles

Thomson discovered the electron e^- in 1897 and gave birth to particle physics.¹ Later in 1911, Rutherford discovered that atoms contain a massive positive nucleus.² As should be familiar the nucleus of hydrogen is the proton p^+ . The last of the three “classical particles” was found in 1932 when Chadwick discovered the neutron n .

2.2. The photon

Planck quantised electromagnetic radiation in 1900 by assuming it came in small packages or “quanta” of energy $\hbar\omega$.³ Later in 1905, Einstein explained the photoelectric effect by assuming the electromagnetic field was quantised. This idea was accepted in 1923 when Compton showed that light scattered from a particle with mass m was shifted in wavelength

$$\lambda' = \lambda + \underbrace{\frac{h}{mc}}_{\lambda_c} (1 - \cos \theta).$$

This formula can be derived by assuming light is a particle with mass $m_\gamma = 0$ and energy $\hbar\omega$. We call this particle the photon γ .

2.3. Mesons

The model above has a problem, since it does not explain why the nucleus is held together. We need a new short-ranged force, which we call the strong force. Yukawa suggested in 1934 that this force occurs due to the exchange of a particle called the pion π . The pion has $m_\pi \sim 300m_e \sim \frac{1}{6}m_p$ and is rather heavy.

$$\text{lepton} \sim e^-$$

$$\text{meson} \sim \pi$$

¹With $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$ we can determine sign of q , the magnitude of v , and the ratio q/m .

²By firing α -particles onto a gold foil.

³With no physical motivation.

$$\text{baryon} \sim p^+, n$$

Later in 1947, the pion π and muon⁴ μ is found in cosmic rays by Powell. Now, we know there are actually three pions: π^0 and $\pi^\pm$.

2.4. Antiparticles

Dirac predicted the existence of antiparticles in 1930. Later in 1931, the positron e^+ was found by Anderson.⁵ Some particles are their own antiparticles, with the photon being an example $\bar{\gamma} = \gamma$. Now, consider the process

$$A + B \rightarrow C + D,$$

which we assume is known to occur. Then by “crossing symmetry” the following processes are also allowed⁶

$$A \rightarrow \bar{B} + C + D$$

$$A + \bar{C} \rightarrow \bar{B} + D$$

$$\bar{C} + \bar{D} \rightarrow \bar{A} + \bar{B}.$$

However, some could be forbidden kinematically. The “reverse” process

$$C + D \rightarrow A + B$$

is also allowed.⁷

2.5. Neutrinos

Consider β -decay as in

$$A \rightarrow B + e^-.$$

We require $Q_A = Q_B - 1$ due to charge conservation. Now, we know $n \in A$ and $p^+ \in B$.

β -decay in the above form is problematic since the electron has too little energy. Pauli resolved this in 1930 by proposing the existence of the neutrino⁸ ν changing β -decay as

⁴“heavy electron”

⁵We have also found μ^+ , p^- , $\bar{n}$ etc.

⁶e.g. $\gamma + e^- \rightarrow \gamma + e^-$ is “the same as” $e^- + e^+ \rightarrow 2\gamma$

⁷This follows from the principle of “detailed balance”.

⁸and antineutrino $\bar{\nu}$

$$n \rightarrow p^+ + e^- + \nu/\bar{\nu},$$

where the neutrino is neutral and interacts weakly, thereby making detection hard.

Likewise, the neutrino was proposed to be involved in other decays⁹

$$\pi \rightarrow \mu + \nu/\bar{\nu}$$

$$\mu \rightarrow e^- + 2\nu/\bar{\nu}.$$

We determine whether a process involves neutrinos or antineutrinos by assigning particles

a “lepton number”¹⁰

$$L(e^-, \mu^-, \nu) = +1, \quad L(e^+, \mu^+, \bar{\nu}) = -1, \quad L(\text{other}) = 0,$$

and proposing lepton number conservation.¹¹ Then

$$n \rightarrow p^+ + e^- + \bar{\nu}$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}$$

$$\pi^+ \rightarrow \mu^+ + \nu$$

$$\mu^- \rightarrow e^- + \nu + \bar{\nu}$$

$$\mu^+ \rightarrow e^+ + \nu + \bar{\nu}.$$

However, the process

$$\mu^- \rightarrow e^- + \gamma$$

never occurs even though no rules seem to be broken. This implies lepton number conservation

of within each generation and the existence of multiple neutrinos. We call these the

electron-neutrino ν_e and muon-neutrino ν_μ . The lepton number is assigned as before, but

now we have an electron number L_e and muon number L_μ . Then

$$n \rightarrow p^+ + e^- + \bar{\nu}_e$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

⁹When the energy of the muon or electron varies there are multiple neutrinos.

¹⁰We can use this as the defining difference between neutrinos and antineutrinos.

¹¹We also have baryon number conservation.

The neutrino was found in 1955 using inverse β -decay

$$\bar{\nu}_e + p^+ \rightarrow n + e^+$$

Later in 1962, the existence of two neutrino types was shown using processes like¹²

$$\bar{\nu}_\mu + p^+ \rightarrow n + \mu^+$$

$$\bar{\nu}_\mu + p^+ \rightarrow n + e^+,$$

with the second never occurring.

2.6. Strangeness

The kaon K^0 was found in 1947 decaying as

$$K^0 \rightarrow \pi^+ + \pi^-.$$

Later in 1949, another kaon K^+ was found decaying as

$$K^+ \rightarrow 2\pi^+ + \pi^-.$$

The K^+ being a charged K^0 was realised in 1956.¹³ The kaons are mesons like pions.¹⁴

The Λ was found in 1950 decaying as

$$\Lambda \rightarrow p^+ + \pi^-.$$

Λ is a baryon by baryon number conservation.¹⁵

These mesons and baryons were “strange” particles, since they are produced in large amounts at a time scale $t \sim 10^{-23}$ s and decay slowly at a time scale $t \sim 10^{-10}$ s. Pais proposed their production and decay mechanisms differed.¹⁶ Now, we would say they are produced by strong interactions and decay by weak interactions. This led to Gell-Mann and Nishijima assigning each particle a “strangeness” in 1953, which is conserved in strong and electromagnetic interactions.

$$S(p, n) = 0, \quad S(\Sigma, \Lambda) = -1, \quad S(\Xi^-, \Xi^0) = -2$$

¹²With $\bar{\nu}_\mu$ from π^- -decay.

¹³We also have $\bar{K}^0, K^-$ etc.

¹⁴We have since found many more mesons: $\rho^0, \rho^+, \rho^-, \eta, \omega, \phi$ etc.

¹⁵We have since found many more baryons: $\Sigma^0, \Sigma^+, \Sigma^-, \Xi^-, \Xi^0, \Delta^-, \Delta^0, \Delta^+, \Delta^{++}$ etc.

¹⁶and “associated production”.

$$S(\eta, \pi s) = 0, \quad S(K^0, K^+) = 1, \quad S(\bar{K}^0, K^-) = -1$$

We know “strangeness” is violated in weak interactions since we have decays like

$$\Lambda \rightarrow p^+ + \pi^-$$

$$\Sigma^+ \rightarrow p^+ + \pi^0.$$

These particles can be organised according to “The Eightfold Way” as shown in Figure 1, Figure 2 and Figure 3.¹⁷

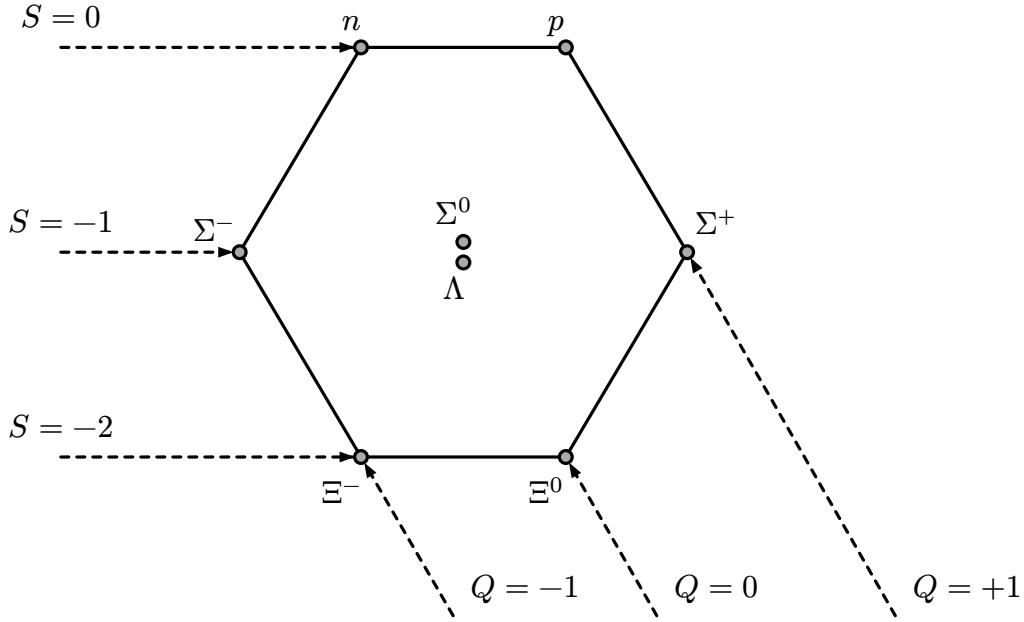


Figure 1: The baryon octet.

¹⁷We have three quarks $\{u, d, s\}$ we have $3 \otimes 3 \otimes 3 = 10 \otimes 8 \otimes 8 \otimes 1$. The octet is spin- $\frac{1}{2}$, while the decuplet is spin- $\frac{3}{2}$.

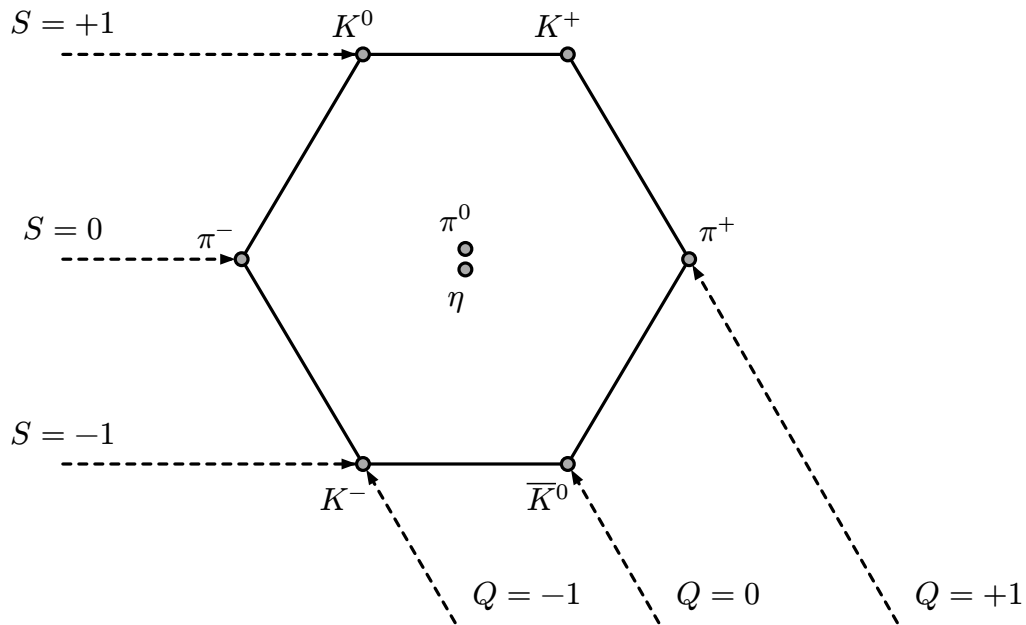


Figure 2: The meson octet.

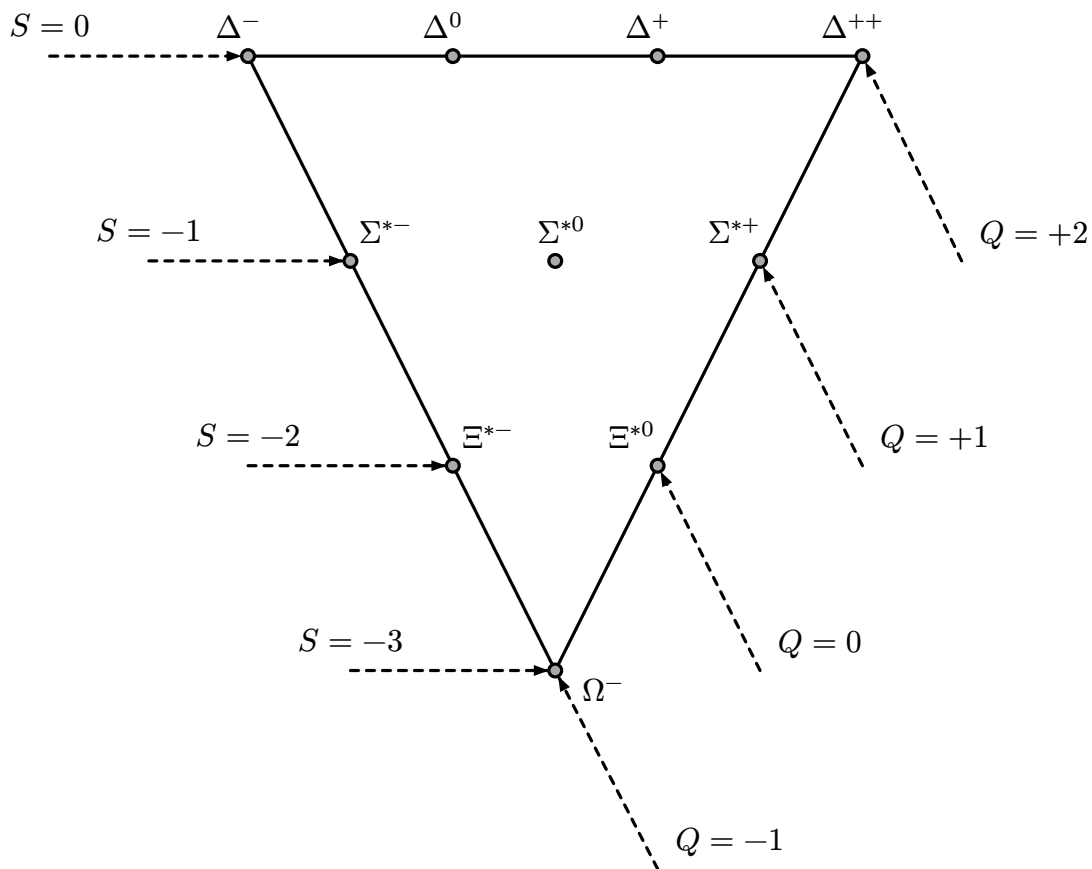


Figure 3: The baryon decuplet.

2.7. Quarks

Gell-Mann and Zweig proposed hadrons were made of quarks q in 1964 to explain the patterns of The Eightfold Way. We have three “flavors” which can be arranged as in Figure 4.¹⁸

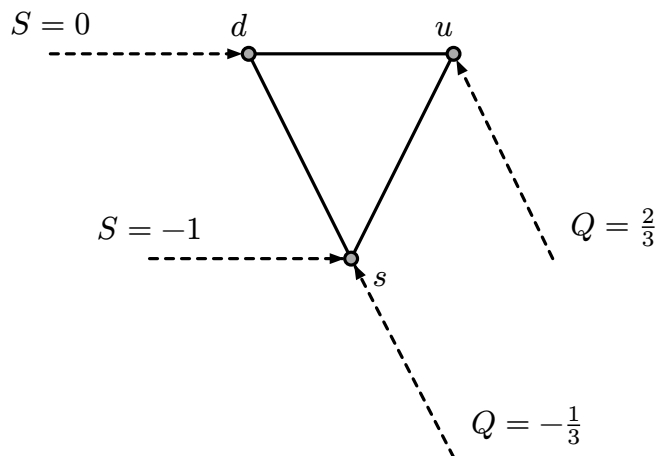


Figure 4: The quarks.

All baryons are composed of three quarks, while all mesons are composed of a quark and an anti-quark

$$B = qqq$$

$$M = q\bar{q}$$

With three quarks we can construct ten baryons and nine mesons.

There are two problems with quarks. The first problem is that we have never observed a quark. The only experimental evidence we have is that the proton has substructure indicating the existence of quarks. The second is that they appear to violate the Pauli exclusion principle since e.g. Δ^{++} contains three u with spin- $\frac{1}{2}$. To solve this Greenberg proposed quarks carry an additional degree of freedom. We call this color and quarks can be red, green or blue. All particles we see in nature must be colorless meaning they have zero color or equal amounts of each.

¹⁸The antiquarks are “upside-down”.

2.8. The Standard Model

As seen in the previous section a bunch of particles were found. The basic building blocks being leptons and quarks

quarks: u, d, c, s, t, b

leptons: $e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau$

With those not mentioned previously eventually being found.

The weak interaction is mediated by the intermediate vector bosons W^+/W^- and Z , the electromagnetic interaction is mediated by photons γ , and strong interaction is mediated by eight gluons.

The final particle found in 2012 was the Higgs boson which is responsible for particles having mass!

3. Dynamics

As far as we know nature is governed by four fundamental forces: electromagnetic, strong, weak, and gravitational. We only consider the first three.

3.1. QED

Quantum electrodynamics is the theory that describes the electromagnetic force and is by far the most simple and best understood of the aforementioned theories.

All electromagnetic phenomena are reducible to the process in Figure 5.¹⁹

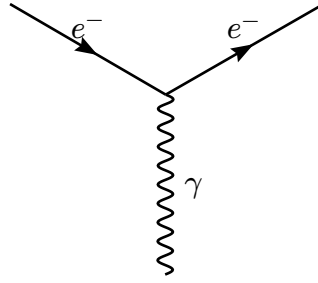


Figure 5: A charged particle e^- enters and emits or absorbs a photon γ before exiting. This is an example of a Feynman diagram. Here time increases towards the right, and there is a “fermion flow”. We see charge is conserved.

This is the only allowed vertex in QED and by patching these processes together we get more complicated interactions. An example is Møller scattering (e^-e^- scattering) seen in Figure 6.

¹⁹Here, we use e^- as a placeholder “charged particle”. We could have $u \rightarrow u + \gamma$.

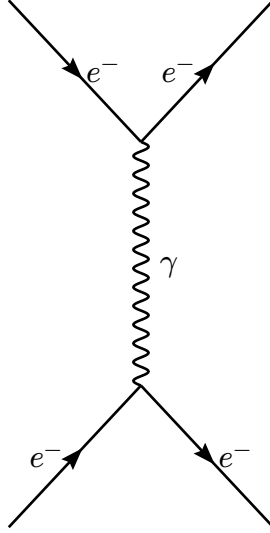


Figure 6: Feynman diagram for Møller scattering. Two electrons e^- scatter by exchanging a photon γ . This is an example of a tree diagram, diagrams with “loops” are called loop diagrams.

Another example is Bhabha scattering (e^-e^+ scattering) seen in Figure 7.

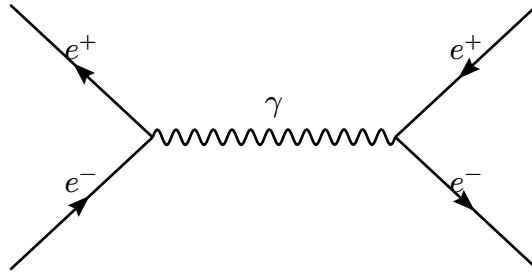


Figure 7: Feynman diagram for Bhabha scattering. An electron e^- and positron e^+ annihilate to form a photon γ which produces a new electron-positron pair. Here, antiparticles (e^+) travel “backward” in time.

This is just the diagram for Møller scattering on its side! We interpret the particles travelling “backward in time” as antiparticles. A second diagram also contributes to Bhabha scattering. This diagram is seen in Figure 8.

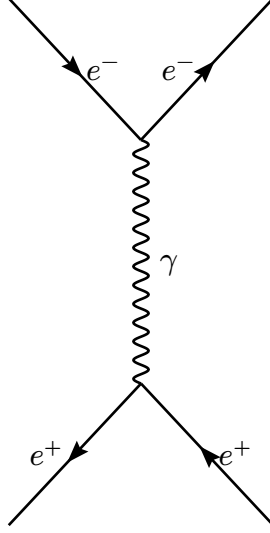


Figure 8: Feynman diagram for Bhabha scattering. An electron e^- and positron e^+ scatter by exchanging a photon γ .

As a final example we consider Compton scattering ($e^-\gamma$ scattering) seen in Figure 9.

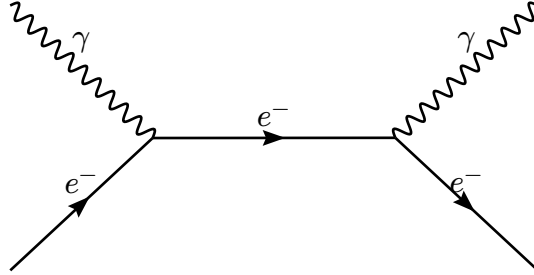


Figure 9: Feynman diagram for Compton scattering.

Almost identical diagrams can be made for pair annihilation and pair production.

We could allow additional vertices in the above to get more complicated diagrams. However, each additional vertex comes with a factor of $\alpha = e^2/\hbar c \sim \frac{1}{137}$ being the fine-structure constant meaning diagrams with additional vertices are less important. The observed process is only represented by external lines. The internal lines represent “virtual particles” and describe the responsible mechanism. The Feynman diagram is a purely symbolic tool! We compute things from Feynman diagrams using Feynman rules which we will discuss later. These rules enforce energy and momentum conservation at each vertex.²⁰

²⁰This implies the QED vertex itself cannot represent a process, i.e. $e^- \rightarrow e^- + \gamma$ and $e^- + e^+ \rightarrow \gamma$ are unphysical.

3.2. QCD

Quantum chromodynamics is the theory that describes the strong force. Now, in QCD color plays the role of charge and the fundamental process is seen in Figure 10.

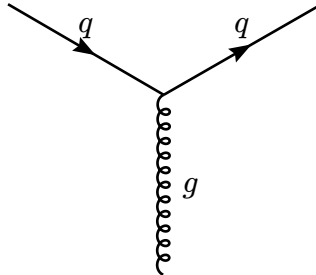


Figure 10: Fundamental vertex for QCD. Here, flavor is conserved.

We can again patch these together to form more complicated interactions. An example is seen in Figure 11.

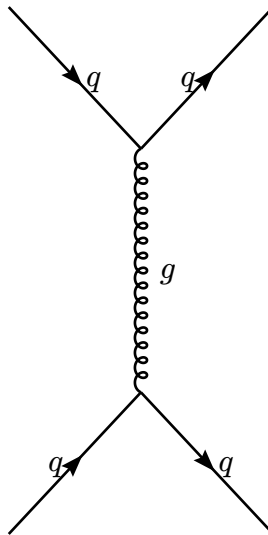


Figure 11: Feynman diagram for quark binding.

We have three types of color and quarks²¹ can change color with the difference being carried by gluons, as seen in Figure 12.

²¹Note, leptons are colorless i.e. do not experience strong interactions.

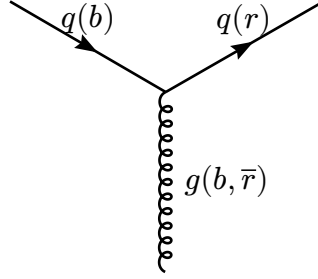


Figure 12: Conversion of a blue quark into a red quark with the help of a “bicolored gluon”. This implies the existence of nine different gluons. However, there are only eight. We see color is conserved.

Now, since gluons are colored they can couple with themselves.²² We get two new kinds of vertices as seen in Figure 13.

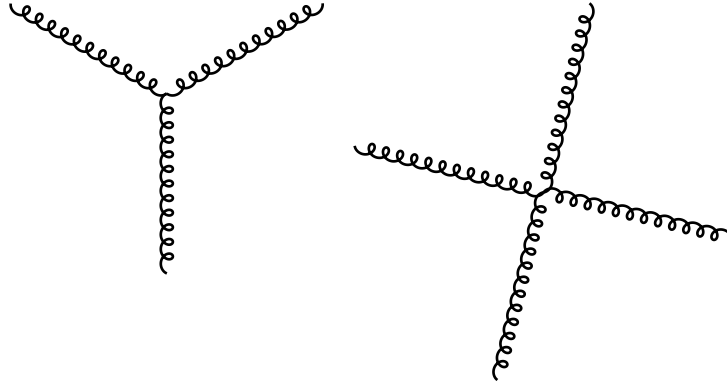


Figure 13: Two additional kinds of fundamental vertices in QCD. Here, the color is also conserved.

This additional coupling necessarily makes QCD way more complicated than QED.

With these vertices one can show $pp \rightarrow pp$ (with $p = uud$) through the exchange of a pion $\pi^0 = u\bar{u}$.

3.3. Weak interactions

All fermions (quarks and leptons) carry a “weak charge” meaning they can all interact through weak interactions. Now, we distinguish between charged interactions mediated by $W^\pm$ and neutral interactions mediated by Z .

The fundamental neutral vertex is seen in Figure 14.

²²Unlike the photon, which is neutral.

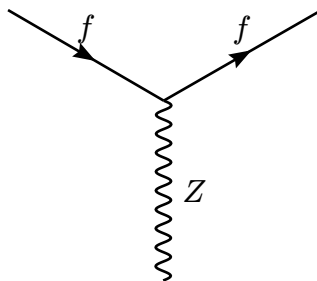


Figure 14: The fundamental neutral vertex. Here, f is a fermion.

Note, that for any process mediated by γ there is an analogous process mediated by Z .

An example of a more complicated interaction is seen in Figure 15.

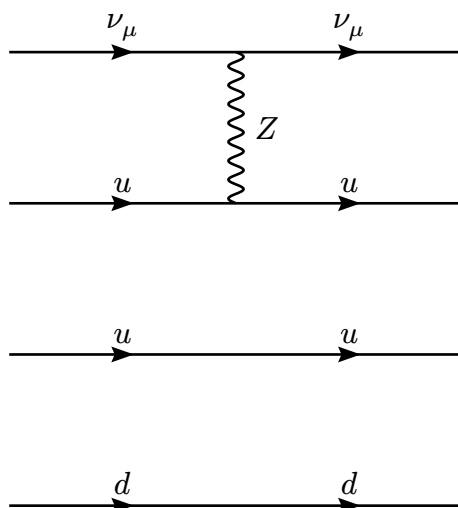


Figure 15: The scattering of a muon-neutrino ν_μ with a proton p . Here, the two up-quarks u are called spectator quarks.

The fundamental charged vertices are seen in Figure 16.

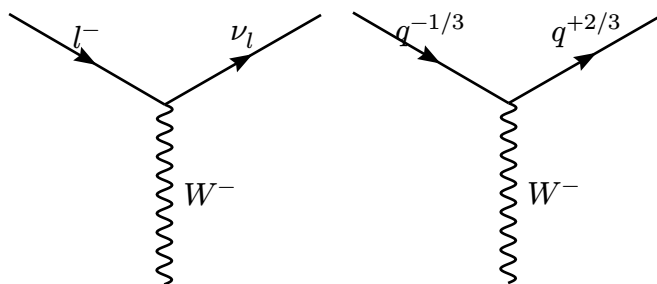


Figure 16: The fundamental charged vertices. The opposite direction is mediated by W^+ . The quarks in the second diagram have the same color but different flavors.

An example of a very important process is seen in Figure 17

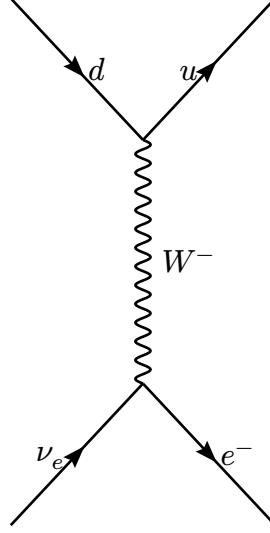


Figure 17: The semileptonic process $d + \nu_e \rightarrow u + e^-$.

This diagram gives us β -decay $n \rightarrow p + e^- + \bar{\nu}_e$! Since $udd \rightarrow uud$ with two spectator quarks. We also have $\mu^- + \nu_e \rightarrow e^- + \nu_\mu$ implying $\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e$.

Now, what we have done so far implies weak interactions only couple pairs (u, d) . However, weak interactions actually couple pairs (u, d') with²³

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

Which we call the CKM-mechanism and accounts for the decay of $\Lambda \rightarrow p^+ + \pi^-$ and $\Omega^- \rightarrow \Lambda + K^-$.²⁴

The $W^\pm$ and Z also couple directly, and since the $W^\pm$ are charged they also couples to photons γ . These interactions are all shown in Figure 18.

²³Also, (c, s') and (t, b') .

²⁴Note, this allows the construction of “penguin diagrams”.

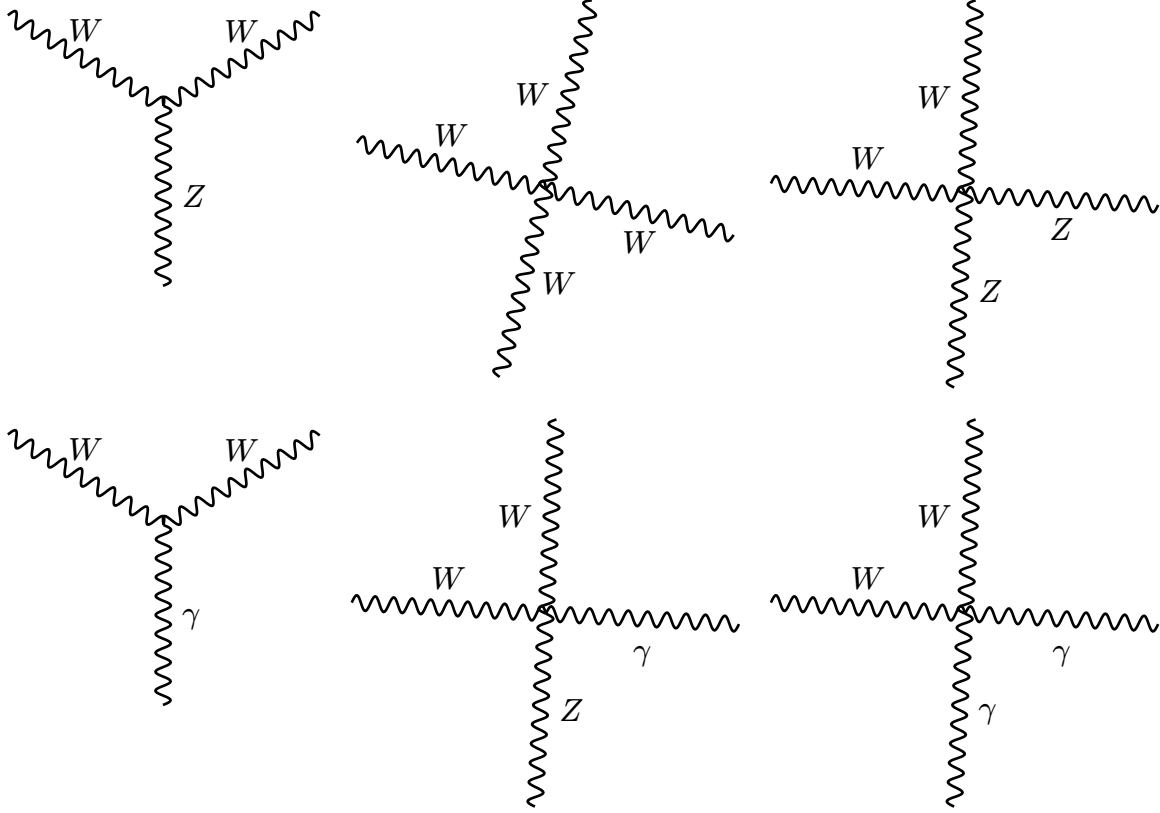


Figure 18: The couplings between $W^\pm$, Z and γ .

3.4. Decays and laws

All particles want to disintegrate into some lighter particles unless some law prevents it. The photon is stable since $m_\gamma = 0$ while the electron is stable since it is the lightest charged particle so by charge conservation it cannot decay. Likewise, the proton is stable due to baryon number conservation. However, most particles want to decay and most “exotic” particles only exist in relatively small amounts.

Any decay is governed by one of the three interactions described above. Some examples are

$$\Delta^{++} \rightarrow p + \pi^+ \quad \text{strong interaction}$$

$$\pi^0 \rightarrow \gamma + \gamma \quad \text{electromagnetic interaction}$$

$$\Sigma^- \rightarrow n + e^- + \bar{\nu}_e \quad \text{weak interaction}$$

Some interactions are less obvious

$$\Sigma^- \rightarrow n + \pi^- \quad \text{weak interaction}$$

$$\Delta^- \rightarrow n + \pi^- \quad \text{strong interaction}$$

where in the Σ^- -decay flavor changes.

We have seen all fundamental vertices and by seeing what is conserved in these we can figure out what is conserved in general. We have conservation of

1. charge.
2. color.
3. baryon number (number of quarks).
4. lepton number.
5. flavor in strong and electromagnetic.

3.5. The Higgs particle

We mention the fundamental vertices of the Higgs particle for completeness. The Higgs particle is neutral and can couple with itself as in Figure 19.

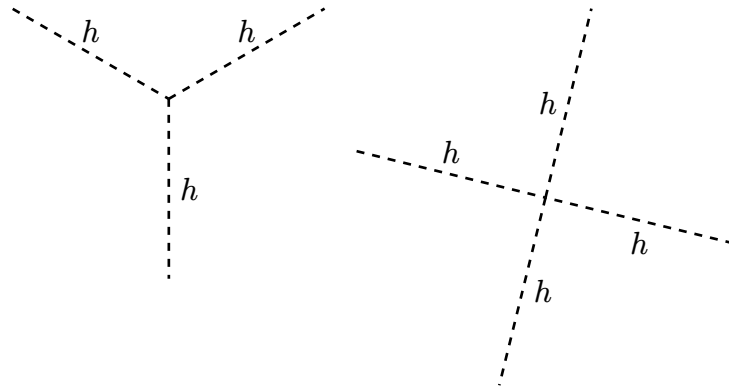


Figure 19: Self-coupling of the Higgs particle.

The Higgs also couples to $W^\pm$ and Z as in Figure 20.

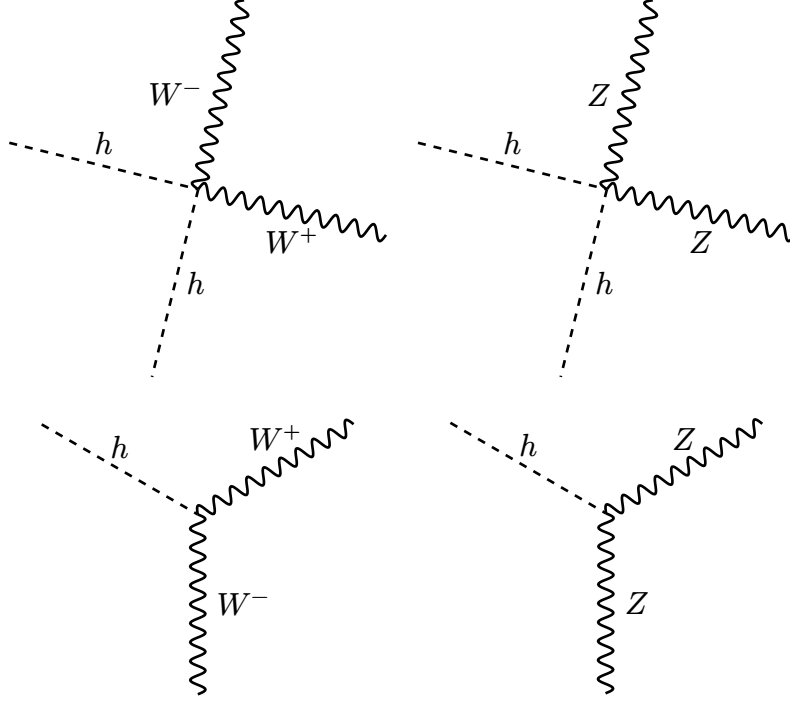


Figure 20: The couplings between $W^\pm, Z$ and h .

We also have an interaction with fermions as in Figure 21.

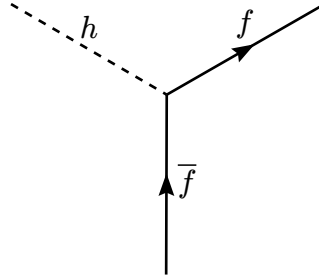


Figure 21: Coupling between fermions and the Higgs particle.

We care about the Higgs since the interaction in Figure 21 is dependent on m_f implying the mass of a given particle is dependent on how much they interact with the Higgs.²⁵

²⁵The Higgs can be created by “gluon fusion” $2g \xrightarrow{t \text{ loop}} h$, we can then observe decays $h \xrightarrow[t \text{ loop}]{W \text{ loop}} 2\gamma$.

4. Special Relativity

4.1. The Lorentz transformations

According to special relativity the laws of physics are valid in all inertial frames.²⁶ Now, consider two inertial frames S and S' , with S' having a uniform velocity $\mathbf{v}$ relative to S .

With coordinates where $\mathbf{v}$ is along $\hat{x}$ these frames are related by

$$x' = \gamma(x - \beta ct); \quad y' = y; \quad z' = z; \quad ct' = \gamma(ct - \beta x),$$

with

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}, \quad \beta = \frac{v}{c}.$$

We call these the Lorentz transformations.²⁷

4.2. Four-vectors

We define the position four-vector x^μ

$$x^\mu = (ct, x, y, z)$$

By comparison with above the Lorentz transformations become

$$x'^0 = \gamma(x^0 - \beta x^1), \text{ etc.}$$

We write $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ ²⁸ where $\Lambda^\mu{}_\nu$ represents the Lorentz transformations for a given $\mathbf{v}$.

We define a general four-vector A^μ as an object transforming like x^μ

$$A^\mu = \Lambda^\mu{}_\nu A^\nu$$

We would like invariant quantities. These can be found by contractions²⁹ using the metric

$g_{\mu\nu}$

$$\text{scalar} = g_{\mu\nu} A^\mu A^\nu \equiv A_\mu A^\mu \equiv A^2$$

We work in flat Minkowski spacetime where $g_{\mu\nu} = \eta_{\mu\nu}$.³⁰ The spacetime interval is the most important example

²⁶By inertial we mean non-accelerating.

²⁷The inverse transformations are given by $\beta \rightarrow -\beta$

²⁸Using the summation convention.

²⁹All indices cancel.

³⁰We use $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

We can extend the definition of a four-vector to define general second-rank tensors $T^{\mu\nu}$ as objects transforming like

$$T'^{\mu\nu} = \Lambda^\mu{}_\kappa \Lambda^\nu{}_\sigma T^{\kappa\sigma}$$

The definition of higher rank tensors is obvious. We can raise and lower indices as implied before using $g_{\mu\nu}$

$$T^\mu{}_\nu = g_{\nu\kappa} T^{\mu\kappa} \sim \text{mixed}$$

We can also use $g_{\mu\nu}$ to take the trace

$$T^\mu{}_\mu = g_{\mu\nu} T^{\mu\nu} \sim \text{scalar}$$

4.3. The four-momentum

We define the proper time τ by

$$d\tau = \frac{dt}{\gamma}$$

This is the time measured in the instantaneous rest-frame.³¹

We define velocity as

$$\mathbf{v} \equiv \frac{d\mathbf{x}}{dt} \sim \text{lab frame}$$

Then the proper velocity is defined by

$$\boldsymbol{\eta} \equiv \frac{d\mathbf{x}}{d\tau} = \gamma \mathbf{v}$$

This quantity is useful since $d\tau$ is a scalar. This motivates the four-velocity η^μ

$$\eta^\mu \equiv \frac{dx^\mu}{d\tau}$$

The related invariant is $\eta_\mu \eta^\mu = c^2$. We can now define the four-momentum p^μ

$$p^\mu \equiv m\eta^\mu$$

With $E \equiv \gamma mc^2$ we can write p^μ as

³¹We use $ds^2 = c^2 d\tau^2 = c^2 dt^2 - d\mathbf{r}^2$.

$$p^\mu = \left(\frac{E}{c}, \mathbf{p} \right)$$

with $\mathbf{p} = \gamma m \mathbf{v}$. The related invariant is

$$p_\mu p^\mu = \frac{E^2}{c^2} - \mathbf{p}^2 \stackrel{\text{rest frame}}{=} m^2 c^2$$

We can expand E to obtain

$$E = \underbrace{mc^2}_{\text{rest energy}} + \underbrace{\frac{1}{2}mv^2 + \dots}_{\text{kinetic energy}}$$

When we consider particles with $m = 0$ we have $p_\mu p^\mu = 0$ implying

$$E = |\mathbf{p}|c$$

We care about p^μ since it is always conserved in any physical process!

4.4. Examples

Consider two lumps with mass m colliding at $\frac{3}{5}c$. After the collision the lumps stick together.

We obviously have $\mathbf{p}_1 = -\mathbf{p}_2$ meaning $\mathbf{p}_M = 0$. Then since $E_1 + E_2 = E_M$ we have

$$Mc^2 = 2E_1 = \frac{2mc^2}{\sqrt{1 - \left(\frac{3}{5}\right)^2}} = \frac{5}{2}mc^2$$

implying

$$M = \frac{5}{2}m > 2m$$

A particle of mass M at rest decays into two pieces with mass m .

By $\mathbf{p}_M = \mathbf{p}_1 + \mathbf{p}_2 = 0$ we have $\mathbf{p}_1 = -\mathbf{p}_2$. Then since $E_M = 2E_1$ we have

$$M = \frac{2m}{\sqrt{1 - \beta^2}} \Rightarrow \beta^2 = 1 - \left(\frac{2m}{M} \right)^2$$

We need $M \geq 2m$ for the process to make sense. We refer to $M = 2m$ as the threshold.³²

A π at rest decays into $\mu + \nu$.

We have

³²When $\beta = 0$.

$$p_\pi = p_\mu + p_\nu \Rightarrow p_\nu = p_\pi - p_\mu$$

implying

$$p_\nu^2 = p_\pi^2 + p_\mu^2 - 2p_\pi \cdot p_\mu$$

Where $p_\nu^2 \sim 0$ and

$$p_\pi^2 = m_\pi^2 c^2; \quad p_\mu^2 = m_\mu^2 c^2; \quad p_\pi \cdot p_\mu = \frac{E_\pi E_\mu}{c^2}$$

Then

$$0 = m_\pi^2 c^2 + m_\mu^2 c^2 - 2m_\pi E_\mu \Rightarrow E_\mu \simeq \frac{m_\pi^2 + m_\mu^2}{2m_\pi} c^2$$

Similarly with $p_\mu = p_\pi - p_\nu$ we find

$$m_\mu^2 c^2 = m_\pi^2 c^2 - 2m_\pi E_\nu$$

using $E_\nu \simeq |\mathbf{p}_\mu|c$ we have

$$|\mathbf{p}_\mu| \simeq \frac{m_\pi^2 - m_\mu^2}{2m_\pi} c$$

A p collides with a p at rest by $p + p \rightarrow p + p + p + \bar{p}$. What is the threshold energy?

We define the center-of-momentum frame to be the frame where $\mathbf{p}_{\text{total}} = 0$. We consider p_{total}^μ before the collision in the lab-frame

$$p_{\text{total}}^\mu = \left(\frac{E + mc^2}{c}, |\mathbf{p}|, 0, 0 \right)$$

and the $p_{\text{total}}'^\mu$ after the collision in the CM-frame

$$p_{\text{total}}'^\mu = (4mc, \mathbf{0}) \sim 4p \text{ at rest}$$

Then we use $p_\mu p^\mu = p'_\mu p'^\mu$ to obtain

$$E = 7mc^2$$

Consider some complicated frame S with

$$E_{\text{total}} = \sum_i \gamma_i m_i c^2; \quad \mathbf{p}_{\text{total}} = \sum_i \gamma_i m_i \mathbf{v}_i$$

Then

$$|\mathbf{p}'_{\text{total}}| = \gamma \left(|\mathbf{p}_{\text{total}}| - \beta \frac{E_{\text{total}}}{c} \right)$$

The CM-frame S' is then given by

$$\beta = \frac{|\mathbf{p}_{\text{total}}|c}{E_{\text{total}}}$$

We always have $\beta < 1$ meaning this frame always exists.

5. Discrete Symmetries

5.1. Angular momentum

Within quantum mechanics we care about the orbital angular momentum $\mathbf{L}$ and the spin³³ angular momentum $\mathbf{S}$. These angular momenta $\mathbf{J}$ are very similar and formally they are essentially the same. However, there are some differences. When performing measurements of $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$ we find these are quantised³⁴

$$L^2 \sim l(l+1)\hbar^2$$

$$L_z \sim m_l \hbar$$

where $l = 0, 1, \dots$ and³⁵

$$m_l = -l, -l+1, \dots, l-1, l$$

When performing measurements of $\mathbf{S}$ we similarly have

$$S^2 \sim s(s+1)\hbar^2$$

$$S_z \sim m_s \hbar$$

where $s = 0, \frac{1}{2}, 1, \dots$ and

$$m_s = -s, -s+1, \dots, s-1, s$$

Now, all particles can be given any l , but each type of particle have their own spin s ! We call particles with half-integer spin fermions and those with integer spin bosons.³⁶

5.2. Addition of angular momentum

We can represent angular momentum states by $|lm_l\rangle$ and $|sm_s\rangle$. We would like to understand composite systems and how to handle the addition of angular momenta

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$$

This could be the total angular momentum of a single particle

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

³³“intrinsic”

³⁴See e.g. Sakurai for details. We measure L^2 since the L_i are non-commuting.

³⁵This implies $\mathbf{L}$ cannot be oriented along $\hat{L}_z$.

³⁶All baryons, leptons, and quarks are fermions, while all mesons and mediators are bosons.

or the spin of a composite system

$$\mathbf{J} = \mathbf{S}_1 + \mathbf{S}_2$$

This is made difficult since

$$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$$

implying we do not have access to all components at once. We are forced to work with a single component J_z and the magnitude J^2 . We would like

$$|jm\rangle \sim |j_1m_1\rangle + |j_2m_2\rangle$$

The J_z add trivially so

$$m = m_1 + m_2$$

However, the J^2 is more annoying and we get all

$$j = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2 - 1, j_1 + j_2$$

These explain the structure of The Eightfold Way since $B = qqq$ and $M = q\bar{q}$ leading to particles with different spins.

5.3. P

Now, we consider the parity operator P which acts on a state $|\mathbf{x}\rangle$ as

$$P|\mathbf{x}\rangle = |-\mathbf{x}\rangle.$$

Lee and Yang proposed³⁷ that weak interactions violate parity around ~ 1956 which was shown experimentally by Wu.³⁸

Now, consider a particle with velocity v along the z -axis (by definition). We define the helicity h as the value of

$$\frac{m_s}{s} = \pm 1,$$

³⁷Motivated by the $\theta\tau$ -puzzle, where two particles seemed identical, but decayed as $\theta \rightarrow 2\pi$ and $\tau \rightarrow 3\pi$, thereby violating parity if θ and τ were the same particle.

³⁸Shown using β -decay of cobalt nuclei. The direction of spin is reversed in a mirror, however, the β -rays are emitted in the same direction. Then reorienting the spin (flipping the image) we find the β -rays to be emitted in the opposite direction in the mirror. Thereby showing β -decay (a weak interaction) is changed under parity.

along this axis.³⁹ When the spin and velocity are parallel ($h = +1$) we call the particle right-handed. Otherwise, we call it left-handed. Under parity we have

$$L \xleftrightarrow{P} R$$

However, the value of h is not invariant!⁴⁰ This is nullified when $v = c$. Now, we assume $m_\nu \sim 0$ implying

$$\nu_L \sim \text{left-handed}$$

$$\bar{\nu}_R \sim \text{right-handed}$$

This can be shown using π^- -decay

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

where μ^- and $\bar{\nu}_\mu$ are emitted back-to-back. The spin is conserved so we can determine the handedness of $\bar{\nu}_\mu$ by measuring μ^- .⁴¹ Likewise, we can use π^+ -decay

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

showing ν_μ is left-handed. This is unlike photons γ which have no preferred handedness, which can be seen from π^0 -decay.

We can characterise objects according to how they transform under parity⁴²

$$\text{scalar} \sim P(s) = s$$

$$\text{pseudoscalar} \sim P(p) = -p$$

$$\text{vector} \sim P(v) = -v$$

$$\text{pseudovector} \sim P(a) = a$$

Also, we have

$$P^2 = \mathbb{1}$$

implying the eigenvalues of P are ± 1 . Now, we will assign

³⁹Here, m_s is the “value” of S_z . An e^- with $s = +\frac{1}{2}$ (along S_z) moving along $+\hat{z}$ would have $h = +1$, while an e^- with $s = -\frac{1}{2}$ moving along $+\hat{z}$ would have $h = -1$. However, an e^- with $s = -\frac{1}{2}$ moving along $-\hat{z}$ would have $h = +1$.

⁴⁰An observer with velocity larger than v would measure $-h$.

⁴¹All μ^- are observed to be right-handed implying all $\bar{\nu}_\mu$ are right-handed. Since $S_\mu + S_\nu \stackrel{!}{=} 0$.

⁴²The “pseudo” implies they transform “unnormally”.

$$P(q) = +1, \quad P(\bar{q}) = -1$$

implying⁴³

$$P(B) = +1, \quad P(\bar{B}) = -1, \quad P(M) = -1$$

To describe photons we use the vector potential A^μ so

$$P(\gamma) = -1$$

5.4. C

Now, we consider the charge conjugation operator C which acts on a state $|p\rangle$ as

$$C|p\rangle = |\bar{p}\rangle.$$

The operator C changes the sign of all internal quantum numbers.⁴⁴ Also, we have

$$C^2 = \mathbb{1},$$

implying the eigenvalues of C are ± 1 . Then an eigenstate of C satisfies

$$C|p\rangle = \pm|p\rangle,$$

which holds for particles who are their own antiparticles (γ, π^0, η etc). We can show a $q\bar{q}$ state with $l > 0$ and spin s is an eigenstate of C with eigenvalue $(-1)^{l+s}$. Then pseudoscalar mesons ($l = 0$ and $s = 0$) have the eigenvalue $(+1)$, while vector mesons ($l = 0$ and $s = 1$) have the eigenvalue (-1) .

Now, consider acting with C on ν_L

$$\nu_L \xrightarrow{C} \bar{\nu}_L \sim \text{does not exist.}$$

This shows weak interactions violate invariance under charge conjugation.

5.5. CP

Consider the π^+ -decay

$$\pi^+ \rightarrow \mu^+ + \nu_{\mu,L}$$

Now, we apply C and P

$$\pi^- \rightarrow \mu^- + \bar{\nu}_{\mu,R}$$

⁴³Note, with angular momentum l we include a factor $(-1)^l$.

⁴⁴This leaves mass, energy, momentum, and spin unchanged.

which does occur, even though C and P individually are violated. However, CP is also violated.

The process

$$K^0 \leftrightarrow \bar{K}^0$$

is possible by weak interactions. When measuring these particles we will then find linear combinations of K^0 and $\bar{K}^0$.

Now, we have

$$P|K^0\rangle = -|K^0\rangle, \quad P|\bar{K}^0\rangle = -|\bar{K}^0\rangle,$$

and

$$C|K^0\rangle = |\bar{K}^0\rangle, \quad C|\bar{K}^0\rangle = |K^0\rangle,$$

implying

$$CP|K^0\rangle = -|\bar{K}^0\rangle, \quad CP|\bar{K}^0\rangle = -|K^0\rangle.$$

We find two eigenstates of CP

$$|K_1\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle),$$

$$|K_2\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle),$$

where

$$CP|K_1\rangle = |K_1\rangle, \quad CP|K_2\rangle = -|K_2\rangle.$$

Now, assuming CP is not violated then K_1 can only decay into states with $CP = +1$, while K_2 can only decay into states with $CP = -1$. This implies⁴⁵

$$K_1 \rightarrow 2\pi, \quad K_2 \rightarrow 3\pi$$

with the K_1 -decay being much quicker.

Now, imagine a K^0 -beam with

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_1\rangle + |K_2\rangle)$$

⁴⁵ 2π has $(+1)(+1)$ and 3π has $(-1)(+1)$, with the factors being PC . Also, K likes decaying into π .

Since the K_1 -decay is quick we will see many 2π -decays initially and many 3π -decays near the end of our beam. The K_1 and K_2 have been found and their lifetimes were determined to be

$$\tau_1 \sim 10^{-10} \text{ sec}$$

$$\tau_2 \sim 10^{-8} \text{ sec}$$

We can then see if CP is violated since we can produce an arbitrarily pure K_2 -beam. We would then still see 2π -decays if CP is violated.⁴⁶

5.6. T

Now, consider the time reversal operator T taking $t \rightarrow -t$. The operator T is important because of the CPT theorem: “any Lorentz invariant local quantum field theory with an Hermitian Hamiltonian $H^\dagger = H$ preserves CPT ”.⁴⁷ This implies T is violated since the combination CPT is preserved. However, this is experimentally very difficult to test.

⁴⁶The “long-lived-species” would contain small amounts of K_1 and would therefore not be an eigenstate of CP as assumed.

⁴⁷The standard model is such a theory.

6. Computing Γ and σ

We can probe interactions through processes like decays and scattering. Now, we would like to be able to compute stuff related to these processes. We essentially need to bridge the gap between experiment and theory.

6.1. Γ

Now, consider a decay

$$A \rightarrow B + C + \dots$$

We can define the decay rate Γ by

$$\Gamma = \text{probability per unit time that particle } A \text{ decays,}$$

or

$$\frac{dN}{dt} = -\Gamma N,$$

implying

$$N = N_0 e^{-t/\tau},$$

where τ is the mean lifetime

$$\tau = \frac{1}{\Gamma}$$

Now, consider multiple decay channels for a particle A . We can then define a total decay rate Γ_{tot} by

$$\Gamma_{\text{tot}} = \sum_i \Gamma_i,$$

where Γ_i is the decay rate of channel i .⁴⁸ We also define the branching ratio by

$$\text{branching ratio for the } i^{\text{th}} \text{ channel} = \frac{\Gamma_i}{\Gamma_{\text{tot}}}$$

Then computing τ and the branching ratios requires the Γ_i .

6.2. σ

⁴⁸Now, $\tau = 1/\Gamma_{\text{tot}}$.

Now, consider a scattering process. We are interested in the cross section σ .⁴⁹

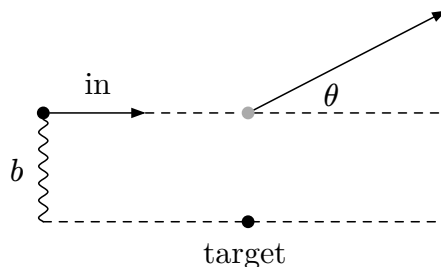


Figure 22: Example scattering.

We consider a particle scattering off a target with scattering angle $\theta(b)$, as shown in Figure 22.⁵⁰ Now, a particle with impact parameter between b and $b + db$ will scatter into an angle between θ and $\theta + d\theta$. More generally, a particle passing through $d\sigma$ will scatter into a solid angle $d\Omega$.

We relate these by

$$d\sigma = D(\theta) d\Omega,$$

where D being the differential cross section.

Now, we would like an expression for $D(\theta)$. We can write

$$\begin{aligned} d\sigma &= \left| \underbrace{\frac{d\phi}{2\pi} 2\pi(b + db)^2}_{\text{outer section}} - \underbrace{\frac{d\phi}{2\pi} \pi b^2}_{\text{inner section}} \right| \\ &= |b db d\phi|, \end{aligned}$$

where we consider the circle of radius b expanding by db and finding the change in $d\sigma$.

Also, we consider small angles $d\phi$ since these lie in the same plane as b whereas θ is “perpendicular”.

⁴⁹Think of σ as the size of our target. We could again define $\sigma_{\text{tot}} = \sum_i \sigma_i$.

⁵⁰Classical hard-sphere scattering gives

$$b \stackrel{\text{hard-sphere}}{=} R \cos \frac{\theta}{2},$$

and Rutherford scattering gives

$$b \stackrel{\text{Rutherford}}{=} \frac{q_1 q_2}{2E} \cot \frac{\theta}{2}.$$

Then⁵¹

$$D(\theta) = \frac{d\sigma}{d\Omega} = \left| \frac{b db d\phi}{\sin \theta d\theta d\phi} \right| = \left| \frac{b}{\sin \theta} \frac{db}{d\theta} \right|$$

The cross section is given by⁵²

$$\sigma = \int d\sigma,$$

all particles within σ will scatter.

6.3. Fermi's Golden Rule

Now, consider a particle decaying

$$1 \rightarrow 2 + 3 + \dots + n.$$

Then Γ is given by Fermi's Golden Rule

$$\begin{aligned} \Gamma = & \frac{S}{2\hbar m_1} \int |\mathcal{M}|^2 (2\pi)^4 \underbrace{\delta^{(4)}(p_1 - p_2 - \dots - p_n)}_{\text{four-momentum conservation}} \\ & \times \prod_{j=2}^n 2\pi \underbrace{\delta(p_j^2 - m_j^2 c^2)}_{\text{all outgoing particles are real}} \underbrace{\Theta(p_j^0)}_{\text{positive energies}} \frac{d^4 p_j}{(2\pi)^4}, \end{aligned}$$

where S is the symmetry factor, for each group of s identical particles S gets a factor of $(1/s!)$. The dynamical information of the process are contained within the amplitude $\mathcal{M}(p_1, \dots, p_n)$ everything else is referred to as phase space (the kinematical information).⁵³

The δ and Θ functions ensure the kinematics make sense. Note, all δ carry (2π) and all dp_j carry $(2\pi)^{-1}$.

We will write $d^4 p_j = dp_j^0 d^3 \mathbf{p}_j$ and

⁵¹We have

$$D \stackrel{\text{hard-sphere}}{=} \frac{R^2}{4} \stackrel{\text{Rutherford}}{=} \left(\frac{q_1 q_2}{4E \sin^2(\theta/2)} \right)^2$$

⁵²We find

$$\sigma \stackrel{\text{hard-sphere}}{=} \pi R^2 \stackrel{\text{Rutherford}}{=} \infty$$

⁵³The units of $\mathcal{M}$ are $(mc)^{4-n}$ where $n = \# \text{external lines}$.

$$\begin{aligned}\delta(p_j^2 - m_j^2 c^2) &= \delta\left((p_j^0)^2 - (\mathbf{p}_j^2 + m_j^2 c^2)\right) \\ &= \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}} \left[\delta\left(p_j^0 - \sqrt{\mathbf{p}_j^2 + m_j^2 c^2}\right) + \delta\left(p_j^0 + \sqrt{\mathbf{p}_j^2 + m_j^2 c^2}\right) \right].\end{aligned}$$

The Θ function kills the spike at $p_j^0 = -\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}$ implying

$$\Theta(p_j^0) \delta(p_j^2 - m_j^2 c^2) = \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}} \delta\left(p_j^0 - \sqrt{\mathbf{p}_j^2 + m_j^2 c^2}\right).$$

Now, we can compute the integral over p_j^0 using the δ function to obtain⁵⁴

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 - \dots - p_n) \prod_{j=2}^n \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}} \frac{d^3 \mathbf{p}_j}{(2\pi)^3},$$

with

$$p_j^0 \rightarrow \sqrt{\mathbf{p}_j^2 + m_j^2 c^2},$$

everywhere.

6.4. Two-particle decay

Now, consider the decay

$$1 \rightarrow 2 + 3$$

Then

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathcal{M}|^2 \frac{\delta^{(4)}(p_1 - p_2 - p_3)}{\sqrt{\mathbf{p}_2^2 + m_2^2 c^2} \sqrt{\mathbf{p}_3^2 + m_3^2 c^2}} d^3 \mathbf{p}_2 d^3 \mathbf{p}_3.$$

We write

$$\delta^{(4)}(p_1 - p_2 - p_3) = \delta(p_1^0 - p_2^0 - p_3^0) \delta^{(3)}(\mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3),$$

and assume $\mathbf{p}_1 = 0$ and $p_1^0 = m_1 c$ giving

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathcal{M}|^2 \frac{\delta\left(m_1 c - \sqrt{\mathbf{p}_2^2 + m_2^2 c^2} - \sqrt{\mathbf{p}_3^2 + m_3^2 c^2}\right)}{\sqrt{\mathbf{p}_2^2 + m_2^2 c^2} \sqrt{\mathbf{p}_3^2 + m_3^2 c^2}} \delta^{(3)}(\mathbf{p}_2 + \mathbf{p}_3) d^3 \mathbf{p}_2 d^3 \mathbf{p}_3$$

⁵⁴Since Θ is unity at $p_j^0 = \sqrt{\mathbf{p}_j^2 + m_j^2 c^2}$.

$$= \frac{S}{32\pi^2\hbar m_1} \int |\mathcal{M}|^2 \frac{\delta\left(m_1c - \sqrt{\mathbf{p}_2^2 + m_2^2c^2} - \sqrt{\mathbf{p}_2^2 + m_3^2c^2}\right)}{\sqrt{\mathbf{p}_2^2 + m_2^2c^2} \sqrt{\mathbf{p}_2^2 + m_3^2c^2}} d^3\mathbf{p}_2.$$

We use spherical coordinates with $d^3\mathbf{p}_2 = r^2 \sin\theta dr d\theta d\phi$ and $r = |\mathbf{p}_2|$

$$\Gamma = \frac{S}{32\pi^2\hbar m_1} \int |\mathcal{M}|^2 \frac{\delta\left(m_1c - \sqrt{r^2 + m_2^2c^2} - \sqrt{r^2 + m_3^2c^2}\right)}{\sqrt{r^2 + m_2^2c^2} \sqrt{r^2 + m_3^2c^2}} r^2 \sin\theta dr d\theta d\phi$$

We had $\mathcal{M} = \mathcal{M}(p_1, p_2, p_3)$ initially and after everything we now have $\mathcal{M} = \mathcal{M}(\mathbf{p}_2)$.

However, $\mathcal{M}$ is a scalar implying $\mathcal{M} = \mathcal{M}(\mathbf{p}_2^2 = r^2)$. We can then do the θ and ϕ integrals

$$\Gamma = \frac{S}{8\pi\hbar m_1} \int_0^\infty |\mathcal{M}(r)|^2 \frac{\delta\left(m_1c - \sqrt{r^2 + m_2^2c^2} - \sqrt{r^2 + m_3^2c^2}\right)}{\sqrt{r^2 + m_2^2c^2} \sqrt{r^2 + m_3^2c^2}} r^2 dr$$

Now, we define

$$u = \sqrt{r^2 + m_2^2c^2} + \sqrt{r^2 + m_3^2c^2}$$

Then

$$\Gamma = \frac{S}{8\pi\hbar m_1} \int_{(m_2+m_3)c}^\infty |\mathcal{M}(r)|^2 \delta(m_1c - u) \frac{r}{u} du$$

Using the δ function we send $u \rightarrow m_1c$ and r to

$$r_0 = \frac{c}{2m_1} \sqrt{m_1^4 + m_2^4 + m_3^4 - 2m_1^2m_2^2 - 2m_1^2m_3^2 - 2m_2^2m_3^2}$$

which is the value of $|\mathbf{p}_2|$ consistent with conservation of energy. Then⁵⁵

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2c} |\mathcal{M}|^2$$

where $|\mathbf{p}|$ is the magnitude of either outgoing momentum.

6.5. Scattering

Now, consider the process

$$1 + 2 \rightarrow 3 + 4 + \dots + n$$

Then σ is given by

⁵⁵Being able to perform all the integrals as above without knowing the form of $\mathcal{M}$ is typically not possible. Also, we assume $m_1 > m_2 + m_3$ otherwise $\Gamma = 0$ as expected.

$$\sigma = \frac{S\hbar^2}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - \dots - p_n) \\ \times \prod_{j=3}^n 2\pi \delta(p_j^2 - m_j^2 c^2) \Theta(p_j^0) \frac{d^4 p_j}{(2\pi)^4}$$

We can do the p_j^0 integrals as before

$$\sigma = \frac{S\hbar^2}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - \dots - p_n) \\ \times \prod_{j=3}^n \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}} \frac{d^3 \mathbf{p}_j}{(2\pi)^3}$$

with

$$p_j^0 = \sqrt{\mathbf{p}_j^2 + m_j^2 c^2}$$

everywhere.

6.6. Two-particle scattering

Now, consider the process

$$1 + 2 \rightarrow 3 + 4$$

We will use the CM-frame where $\mathbf{p}_2 = -\mathbf{p}_1$ and

$$\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2} = \frac{(E_1 + E_2)|\mathbf{p}_1|}{c}$$

Then

$$\sigma = \frac{S\hbar^2 c}{64\pi^2 (E_1 + E_2)|\mathbf{p}_1|} \int |\mathcal{M}|^2 \frac{\delta^{(4)}(p_1 + p_2 - p_3 - p_4)}{\sqrt{\mathbf{p}_3^2 + m_3^2 c^2} \sqrt{\mathbf{p}_4^2 + m_4^2 c^2}} d^3 \mathbf{p}_3 d^3 \mathbf{p}_4$$

We write

$$\delta^{(4)}(p_1 + p_2 - p_3 - p_4) = \delta\left(\frac{E_1 + E_2}{c} - p_3^0 - p_4^0\right) \delta^{(3)}(\mathbf{p}_3 + \mathbf{p}_4)$$

and perform the $\mathbf{p}_4$ integral using the δ function

$$\sigma = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1 + E_2)|\mathbf{p}_1|} \int |\mathcal{M}|^2 \frac{\delta\left((E_1 + E_2)/c - \sqrt{\mathbf{p}_3^2 + m_3^2 c^2} - \sqrt{\mathbf{p}_3^2 + m_4^2 c^2}\right)}{\sqrt{\mathbf{p}_3^2 + m_3^2 c^2} \sqrt{\mathbf{p}_3^2 + m_4^2 c^2}} d^3 \mathbf{p}_3$$

Here, we cannot do the θ and ϕ integrals since $\mathcal{M}$ depends on the direction of $\mathbf{p}_3$ as well as its magnitude.⁵⁶ We instead write

$$d^3 \mathbf{p}_3 = r^2 dr d\Omega$$

with $r = |\mathbf{p}_3|$ implying

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1 + E_2)|\mathbf{p}_1|} \underbrace{\int_0^\infty |\mathcal{M}|^2 \frac{\delta\left((E_1 + E_2)/c - \sqrt{r^2 + m_3^2 c^2} - \sqrt{r^2 + m_4^2 c^2}\right)}{\sqrt{r^2 + m_3^2 c^2} \sqrt{r^2 + m_4^2 c^2}} r^2 dr}_{\text{as before}}$$

We do the integral as before to obtain

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

where $|\mathbf{p}_f|$ and $|\mathbf{p}_i|$ are the magnitudes of either- outgoing and incoming momentum.

6.7. ABC theory

Now, we want to compute amplitudes $\mathcal{M}$. This is done by writing all the relevant Feynman diagrams for a process and then using the Feynman rules. We will consider a toy model which will serve to illustrate the method.

We imagine a world with three particles: (A, B, C) .⁵⁷ We assume they are spin zero and each is its own antiparticle. There is one fundamental vertex which is shown in Figure 23. This diagram contributes to $A \rightarrow B + C$.

⁵⁶We have three invariants $\mathbf{p}_1^2$, $\mathbf{p}_3^2$ and $\mathbf{p}_1 \cdot \mathbf{p}_3 = |\mathbf{p}_1||\mathbf{p}_3| \cos \theta$. However, $\mathbf{p}_1$ is fixed so the only integration variables $\mathcal{M}$ can depend on are $|\mathbf{p}_3|$ and θ .

⁵⁷We will assume $m_A > m_B + m_C$.

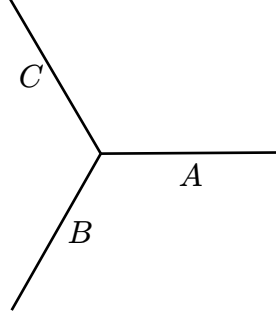


Figure 23: The fundamental vertex in ABC theory.

We will also consider other processes such as $A + A \rightarrow B + B$ and $A + B \rightarrow A + B$.

Now, consider a general Feynman diagram as shown in Figure 24.

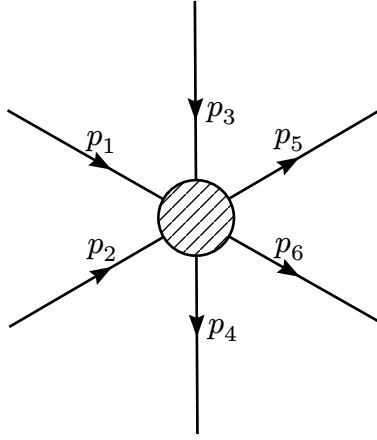


Figure 24: A general Feynman diagram.

The Feynman rules are:

1. Label the incoming- and outgoing momenta $p_1, \dots, p_n$ as above and label internal momenta $q_1, q_2, \dots$
2. Each vertex carries a factor $-ig$. Where g is referred to as the coupling.
3. Each internal line carries a propagator⁵⁸

$$\frac{i}{q_j^2 - m_j^2 c^2}$$

4. Each vertex carries a δ function

$$(2\pi)^4 \delta^{(4)}(k_1 + k_2 + k_3)$$

with k_i being negative for outward momenta.

⁵⁸We have $q_j^2 \neq m_j^2 c^2$ since virtual particles are off-shell.

5. Each internal line carries an integral

$$\int \frac{d^4 q_j}{(2\pi)^4}$$

6. The result will carry a factor

$$(2\pi)^4 \delta^{(4)}(p_1 + p_2 + \dots - p_n)$$

replace this by i .

6.8. ABC examples

Now, consider the decay

$$A \rightarrow B + C$$

The Feynman diagram for this process (see Figure 23) has no internal lines so we immediately obtain

$$\mathcal{M} \sim (-ig)(2\pi)^4 \delta^{(4)}(p_1 - p_2 - p_3)$$

which upon $(2\pi)^4 \delta^{(4)} \rightarrow i$ becomes

$$\mathcal{M} = g + \mathcal{O}(g^2)$$

We can then compute Γ

$$\Gamma = \frac{g^2 |\mathbf{p}|}{8\pi \hbar m_A^2 c}$$

where

$$|\mathbf{p}| = \frac{c}{2m_A} \sqrt{m_A^4 + m_B^4 + m_C^4 - 2m_A^2 m_B^2 - 2m_A^2 m_C^2 - 2m_B^2 m_C^2}$$

We have found the lifetime of A

$$\tau_A = \frac{8\pi \hbar m_A^2 c}{g^2 |\mathbf{p}|}$$

Now, consider the process

$$A + A \rightarrow B + B$$

The Feynman diagram is shown in Figure 25.

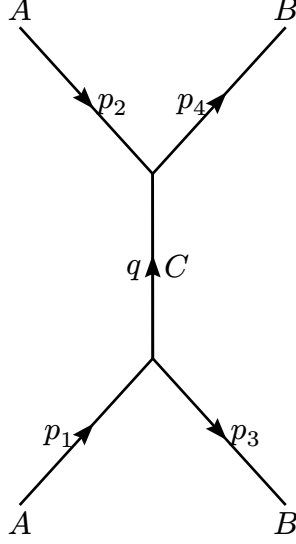


Figure 25: Lowest order diagram for $A + A \rightarrow B + B$.

With the Feynman rules we obtain

$$\begin{aligned}\mathcal{M} &\sim (-ig)^2 \int \frac{d^4 q}{(2\pi)^4} \frac{i}{q^2 - m_C^2 c^2} (2\pi)^4 \delta^{(4)}(p_1 - q - p_3) (2\pi)^4 \delta^{(4)}(p_2 + q - p_4) \\ &\sim -ig^2 (2\pi)^4 \frac{\delta^{(4)}(p_1 + p_2 - p_3 - p_4)}{(p_4 - p_2)^2 - m_C^2 c^2}\end{aligned}$$

which upon $(2\pi)^4 \delta^{(4)} \rightarrow i$ becomes

$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2}$$

We can include a second diagram of order $\mathcal{O}(g^2)$ found by twisting the B lines. This amounts to $p_3 \leftrightarrow p_4$ implying

$$\mathcal{M}_{\text{tot}} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

Now, we can compute $D(\theta)$ in the CM-frame. We assume $m_A = m_B = m$ and $m_C = 0$ implying

$$(p_4 - p_2)^2 - m_C^2 c^2 = p_4^2 + p_2^2 - 2p_2 \cdot p_4 = -2|\mathbf{p}|^2(1 - \cos \theta)$$

$$(p_3 - p_2)^2 - m_C^2 c^2 = p_3^2 + p_2^2 - 2p_3 \cdot p_2 = -2|\mathbf{p}|^2(1 + \cos \theta)$$

with $\mathbf{p}$ being the incoming momentum of particle 1. Then

$$\mathcal{M} = -\frac{g^2}{p^2 \sin^2 \theta}$$

and

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} \left(\frac{\hbar c g^2}{16\pi E p^2 \sin^2 \theta} \right)^2$$

since $S = 1/2$.

7. Quantum Electrodynamics

7.1. The Dirac equation

We can “derive” the Klein-Gordon equation⁵⁹ by considering

$$p^\mu p_\mu - m^2 c^2 = 0$$

and promoting p_μ to an operator

$$p_\mu \rightarrow i\hbar\partial_\mu$$

Then we obtain

$$\partial^\mu \partial_\mu \psi + \frac{m^2 c^2}{\hbar^2} \psi = 0$$

or with $\square \equiv \partial_\mu \partial^\mu$ we have

$$\left[\square + \frac{m^2 c^2}{\hbar^2} \right] \psi = 0$$

which is the Klein-Gordon equation. The Klein-Gordon equation is problematic since the equation is second order in time and only describes spin zero particles.

This lead Dirac to try and find an equation consistent with

$$p^\mu p_\mu - m^2 c^2 = 0,$$

which is first order in time. We “factor” the above

$$p^\mu p_\mu - m^2 c^2 = (\beta^\kappa p_\kappa + mc)(\gamma^\lambda p_\lambda - mc) = \beta^\kappa \gamma^\lambda p_\kappa p_\lambda - mc(\beta^\kappa - \gamma^\kappa)p_\kappa - m^2 c^2$$

We see $\beta^\kappa = \gamma^\kappa$ and

$$p^\mu p_\mu = \gamma^\kappa \gamma^\lambda p_\kappa p_\lambda = \frac{1}{2}(\gamma^\kappa \gamma^\lambda + \gamma^\lambda \gamma^\kappa) p_\kappa p_\lambda \stackrel{!}{=} \eta^{\kappa\lambda} p_\kappa p_\lambda$$

implying the $\gamma^{\mu 60}$ are satisfy the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$$

We will use the following γ^μ

⁵⁹Which describes spin zero particles.

⁶⁰We could write $\gamma_{\alpha\beta}^\mu$ with α, β being spinor indices.

$$\gamma^0 = \begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix},$$

where the σ^i are the Pauli matrices. However, many other γ^μ also work.

We find

$$\gamma^\mu p_\mu - mc = 0,$$

and promoting p_μ to an operator we obtain

$$(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0,$$

which is the Dirac equation. We will call ψ the Dirac spinor. Note, the components of ψ do not transform like a four-vector!

7.2. $\mathbf{p} = 0$ solutions

Now, consider $\psi(\mathbf{x}, t) = \psi(t)$ or $\mathbf{p} = 0$. We find

$$i\hbar\gamma^0\partial_t\psi - mc^2\psi = 0,$$

or

$$\begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix} \begin{pmatrix} \partial_t\psi_A \\ \partial_t\psi_B \end{pmatrix} = -i\frac{mc^2}{\hbar} \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix},$$

implying

$$\partial_t\psi_A = -i\left(\frac{mc^2}{\hbar}\right)\psi_A, \quad \partial_t\psi_B = i\left(\frac{mc^2}{\hbar}\right)\psi_B,$$

which has solutions

$$\psi_A = \psi_A(0) \exp\left[-i\left(\frac{mc^2}{\hbar}\right)t\right], \quad \psi_B = \psi_B(0) \exp\left[+i\left(\frac{mc^2}{\hbar}\right)t\right].$$

Now, consider the usual time-dependence

$$\text{time dependence} \sim \exp\left(-\frac{iEt}{\hbar}\right),$$

implying ψ_A corresponds to a particle with $E = mc^2$. However, ψ_B corresponds to a particle with negative energy $E = -mc^2$! We say these correspond to antiparticles. We have found four independent solutions ($\mathbf{p} = 0$)

$$\psi^{(1)} = \exp \left[-i \left(\frac{mc^2}{\hbar} \right) t \right] \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \sim \text{spin-up } e^-$$

$$\psi^{(2)} = \exp \left[-i \left(\frac{mc^2}{\hbar} \right) t \right] \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \sim \text{spin-down } e^-$$

$$\psi^{(3)} = \exp \left[+i \left(\frac{mc^2}{\hbar} \right) t \right] \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \sim \text{spin-down } e^+$$

$$\psi^{(4)} = \exp \left[+i \left(\frac{mc^2}{\hbar} \right) t \right] \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \sim \text{spin-up } e^+$$

7.3. Plane-wave solutions

Now, consider plane-wave solutions

$$\psi(x) = ae^{-ik^\mu x_\mu} u(k),$$

implying

$$(\hbar\gamma^\mu k_\mu - mc)u = 0.$$

or

$$u_A = \frac{1}{k^0 - mc/\hbar} (\mathbf{k} \cdot \boldsymbol{\sigma}) u_B, \quad u_B = \frac{1}{k^0 + mc/\hbar} (\mathbf{k} \cdot \boldsymbol{\sigma}) u_A.$$

We can find a condition on k_μ by

$$u_A = \frac{(\mathbf{k} \cdot \boldsymbol{\sigma})^2}{(k^0)^2 - (mc/\hbar)^2} u_A \stackrel{(\mathbf{k} \cdot \boldsymbol{\sigma})^2 = k^2}{=} \frac{k^2}{(k^0)^2 - (mc/\hbar)^2} u_A$$

implying⁶¹

⁶¹or the trivial solution $u_A = u_B = 0$

$$k^\mu k_\mu = \left(\frac{mc}{\hbar}\right)^2$$

We find $\hbar k^\mu$ is a four-vector whose square is $m^2 c^2$ implying

$$\hbar k^\mu = \pm p^\mu.$$

We say $\hbar k^\mu = +p^\mu$ corresponds to particle states and $\hbar k^\mu = -p^\mu$ corresponds to antiparticle states.⁶²

We find four solutions:

$$u_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow u_B = \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{p^0 + mc} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{c}{E + mc^2} \begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix}$$

$$u_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow u_B = \frac{c}{E + mc^2} \begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix}$$

$$u_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow u_A = \frac{c}{E + mc^2} \begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix}$$

$$u_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow u_A = \frac{c}{E + mc^2} \begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix}$$

Where $\hbar k^\mu = +p^\mu$ in the first two. We normalise these by

$$u^\dagger u = \frac{2E}{c}.$$

With

$$N \equiv \sqrt{\frac{E + mc^2}{c}},$$

we find the “canonical” solutions

$$u^{(1)} = N \begin{pmatrix} 1 \\ 0 \\ \frac{cp_z}{E+mc^2} \\ \frac{c(p_x+ip_y)}{E+mc^2} \end{pmatrix}, \quad u^{(2)} = N \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x-ip_y)}{E+mc^2} \\ \frac{-cp_z}{E+mc^2} \end{pmatrix}$$

⁶²We have $\sim e^{\mp i k_\mu x^\mu}$ then acting with $p^\mu = i\hbar \partial^\mu$ makes this obvious.

$$v^{(1)} = N \begin{pmatrix} \frac{c(p_x - ip_y)}{E + mc^2} \\ \frac{-cp_z}{E + mc^2} \\ 0 \\ 1 \end{pmatrix}, \quad v^{(2)} = -N \begin{pmatrix} \frac{cp_z}{E + mc^2} \\ \frac{c(p_x + ip_y)}{E + mc^2} \\ 1 \\ 0 \end{pmatrix}$$

Now, we can write the plane-wave solutions⁶³

$$\psi = a \exp \left[-\frac{ip_\mu x^\mu}{\hbar} \right] u \sim \text{particles}$$

$$\psi = a \exp \left[+\frac{ip_\mu x^\mu}{\hbar} \right] v \sim \text{antiparticles}$$

Note, the particle states satisfy

$$(\gamma^\mu p_\mu - mc)u = 0.$$

However, the antiparticle states satisfy

$$(\gamma^\mu p_\mu + mc)v = 0.$$

7.4. Bilinear covariants

We define the adjoint $\bar{\psi}$ by

$$\bar{\psi} \equiv \psi^\dagger \gamma^0$$

With $\bar{\psi}$ we can make scalars

$$\bar{\psi}\psi = \psi^\dagger \gamma^0 \psi \sim \text{scalar}$$

We can also make pseudoscalars by defining γ^5

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3,$$

which satisfies

$$\{\gamma^\mu, \gamma^5\} = 0$$

We can then make

$$\bar{\psi}\gamma^5\psi \sim \text{pseudoscalar}$$

Likewise, we can make more complicated stuff

$$\bar{\psi}\gamma^\mu\psi \sim \text{vector}$$

⁶³With motion along z the $u^{(1)}, v^{(1)}$ are spin up, while the $u^{(2)}, v^{(2)}$ are spin down.

$$\bar{\psi}\gamma^\mu\gamma^5\psi \sim \text{pseudovector}$$

$$\bar{\psi}\sigma^{\mu\nu}\psi \sim \text{antisymmetric tensor}$$

with

$$\sigma^{\mu\nu} \equiv \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$$

All other bilinears will always reduce to one of these.

Note, one can show

$$j^\mu = \bar{\psi}\gamma^\mu\psi$$

is conserved and leads to charge!

7.5. γ and Maxwell's equations

We define the four-potential

$$A^\mu = (V, \mathbf{A})$$

and the four-current

$$J^\mu = (c\rho, \mathbf{J})$$

We can then write Maxwell's equations as

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu$$

where we have introduced the antisymmetric Faraday tensor

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

Note, this implies

$$\partial_\mu J^\mu = 0$$

We see Maxwell's equations are unchanged under gauge transformations

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda$$

We refer to this as having gauge freedom. We can use this to impose

$$\partial_\mu A^\mu = 0$$

which is called the Lorenz gauge. With this gauge Maxwell's equations become

$$\square A^\mu = \frac{4\pi}{c} J^\mu$$

This still does not uniquely specify A^μ since any gauge transformation with

$$\square \lambda = 0$$

will still leave our equations unchanged. We remedy this by imposing

$$A^0 = 0$$

in empty space where $J^\mu = 0$. Then the Lorenz gauge becomes

$$\nabla \cdot \mathbf{A} = 0$$

which is called the Coulomb gauge. Note, now we have broken the manifest Lorentz invariance.

A^μ becomes the wave function of the photon γ in quantum electrodynamics. The free photon γ satisfies the above with $J^\mu = 0$

$$\square A^\mu = 0,$$

which we see is the Klein-Gordon equation for a particle with $m = 0$. Now, consider plane-wave solutions

$$A^\mu(x) = a \exp\left[-\frac{ip_\mu x^\mu}{\hbar}\right] \epsilon^\mu(p),$$

where $p^\mu = (E/c, \mathbf{p})$ and ϵ^μ is the polarization vector.⁶⁴ We find $p^\mu p_\mu = 0$.

With the Lorenz gauge

$$p^\mu \epsilon_\mu = 0,$$

and in the Coulomb gauge

$$\epsilon^0 = 0,$$

implying

$$\boldsymbol{\epsilon} \cdot \mathbf{p} = 0.$$

We find $\boldsymbol{\epsilon}$ is perpendicular to $\mathbf{p}$. We say a free photon γ is transversely polarised. We can always find two $\boldsymbol{\epsilon}$ perpendicular to $\mathbf{p}$. We take $\mathbf{p} = p_z \hat{z}$ then

⁶⁴ ϵ^μ characterises the spin of the photon γ .

$$\epsilon^{(1)} = (1, 0, 0), \quad \epsilon^{(2)} = (0, 1, 0).$$

We find the photon γ has two “spin” states.⁶⁵ So the photon γ can have

$$m_s = \pm s \sim h = \pm 1$$

Note, the linear polarisation $\epsilon^{(1,2)}$ above are superpositions of states with definite h . We could have used

$$\epsilon^{(+)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, \quad \epsilon^{(-)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

7.6. Summary

We have found particles e^- and antiparticles e^+ with $p = (E/c, \mathbf{p})$ are represented by

$$\psi(x) = a \exp\left[-\frac{ip_\mu x^\mu}{\hbar}\right] u^{(s)}(p) \sim e^-$$

$$\psi(x) = a \exp\left[+\frac{ip_\mu x^\mu}{\hbar}\right] v^{(s)}(p) \sim e^+$$

with $s = 1, 2$. We know the spinors $u^{(s)}$ and $v^{(s)}$ satisfy

$$(\gamma^\mu p_\mu - mc)u = 0, \quad (\gamma^\mu p_\mu + mc)v = 0$$

$$\bar{u}(\gamma^\mu p_\mu - mc) = 0, \quad \bar{v}(\gamma^\mu p_\mu + mc) = 0$$

and (orthogonal and normalised)

$$\bar{u}^{(1)}u^{(2)} = 0, \quad \bar{v}^{(1)}v^{(2)} = 0$$

$$\bar{u}u = 2mc, \quad \bar{v}v = -2mc$$

We also have the completeness relations

$$\sum_s u^{(s)}\bar{u}^{(s)} = \gamma^\mu p_\mu + mc, \quad \sum_s v^{(s)}\bar{v}^{(s)} = \gamma^\mu p_\mu - mc$$

with an example being the “canonical” $\{u^{(s)}, v^{(s)}\}$ above.

We have found photons γ with $p = (E/c, \mathbf{p})$ are represented by

⁶⁵Which all massless particles do, except for $s = 0$ which only has one.

$$A_\mu(x) = a \exp\left[-\frac{ip_\mu x^\mu}{\hbar}\right] \epsilon_\mu^{(s)}$$

with $s = 1, 2$.

The $\epsilon_\mu^{(s)}$ satisfy (Lorenz gauge)

$$p^\mu \epsilon_\mu = 0$$

and (orthogonal and normalised)

$$\epsilon_\mu^{(1)*} \epsilon^{(2)\mu} = 0$$

$$\epsilon^{\mu*} \epsilon_\mu = 1$$

Also, with the Coulomb gauge we have

$$\epsilon^0 = 0; \quad \epsilon \cdot p = 0$$

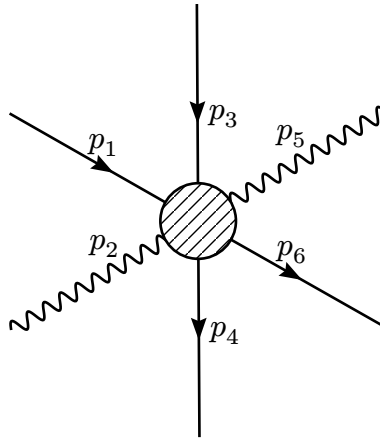
and the ϵ are complete

$$\sum \epsilon_i^{(s)} \epsilon_j^{(s)*} = \delta_{ij} - \hat{p}_i \hat{p}_j$$

with an example being the $\{\epsilon^{(1)}, \epsilon^{(2)}\}$ above.

7.7. Feynman rules for QED

Now, consider a general Feynman diagram in QED



The Feynman rules are:

1. Label the incoming- and outgoing momenta $p_1, \dots, p_n$ as above and label internal momenta $q_1, \dots$ ⁶⁶

⁶⁶Note, the external momenta should go forward in time.

2. Each external line carries a factor:

$$e^- \sim \begin{cases} \text{incoming: } u \\ \text{outgoing: } \bar{u} \end{cases}$$

$$e^+ \sim \begin{cases} \text{incoming: } \bar{v} \\ \text{outgoing: } v \end{cases}$$

$$\gamma \sim \begin{cases} \text{incoming: } \epsilon_\mu \\ \text{outgoing: } \epsilon_\mu^* \end{cases}$$

3. Each vertex carries a factor $ig_e\gamma^\mu$. Where g_e is related to the fundamental charge e by

$$g_e = e\sqrt{\frac{4\pi}{\hbar c}} = \sqrt{4\pi\alpha}$$

4. Each internal line carries a propagator:

$$e^\pm \sim \frac{i(\gamma^\mu q_\mu + mc)}{q^2 - m^2c^2}$$

$$\gamma \sim \frac{-i\eta_{\mu\nu}}{q^2}$$

5. Each vertex carries a δ function

$$(2\pi)^4\delta^{(4)}(k_1 + k_2 + k_3)$$

with k_i being negative for outward momenta.

6. Each internal line carries an integral

$$\int \frac{d^4q}{(2\pi)^4}$$

7. The result will carry a factor

$$(2\pi)^4\delta^{(4)}(p_1 + p_2 + \dots - p_n)$$

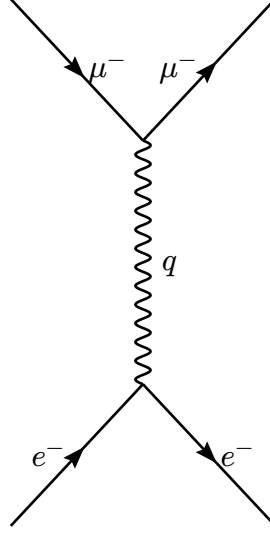
replace this by i .

8. We include a $-$ between diagrams differing only by the interchange of two incoming or outgoing $e^{\pm 67}$

⁶⁷or of an incoming $e^\mp$ and outgoing $e^\pm$

7.8. $e^- \mu^-$ scattering

Now, consider $e^- \mu^-$ scattering. We have one diagram contributing to second-order



with q upwards. We find

$$\mathcal{M} \sim \int \frac{d^4 q}{(2\pi)^4} \overbrace{\left[\bar{u}^{(s_3)}(p_3) i g_e \gamma^\mu u^{(s_1)}(p_1) \right]}^{e^- \text{ line}} \frac{-i \eta_{\mu\nu}}{q^2} \underbrace{\left[\bar{u}^{(s_4)}(p_4) i g_e \gamma^\nu u^{(s_2)}(p_2) \right]}_{\mu^- \text{ line}} \\ \times (2\pi)^4 \delta^{(4)}(p_2 - p_4 + q) (2\pi)^4 \delta^{(4)}(p_1 - p_3 - q)$$

Where the “ e^- line” and “ μ^- line” have been constructed backwards and “connected” with the photon γ propagator. We find

$$\mathcal{M} = \left[\bar{u}^{(s_3)}(p_3) i g_e \gamma^\mu u^{(s_1)}(p_1) \right] \frac{\eta_{\mu\nu}}{(p_1 - p_3)^2} \left[\bar{u}^{(s_4)}(p_4) i g_e \gamma^\nu u^{(s_2)}(p_2) \right] \\ = -\frac{g_e^2}{(p_1 - p_3)^2} \left[\bar{u}^{(s_3)}(p_3) \gamma^\mu u^{(s_1)}(p_1) \right] \left[\bar{u}^{(s_4)}(p_4) \gamma_\mu u^{(s_2)}(p_2) \right]$$

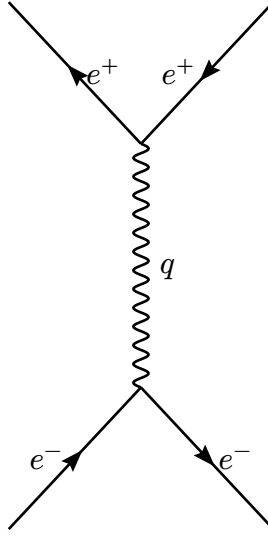
7.9. $e^- e^-$ scattering

Now, consider $e^- e^-$ scattering. We have the same diagram as above contributing. However, we also have a second diagram where the e^- emerging with (p_3, s_3) comes from the e^- with (p_2, s_2) instead of the e^- with (p_1, s_1) . We essentially “twist” the diagram above. We find

$$\mathcal{M} = \underbrace{\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3)\gamma^\mu u(1)] [\bar{u}(4)\gamma_\mu u(2)]}_{\text{by rule 8}} + \underbrace{\frac{g_e^2}{(p_1 - p_4)^2} [\bar{u}(4)\gamma^\mu u(1)] [\bar{u}(3)\gamma_\mu u(2)]}_{3 \leftrightarrow 4}$$

7.10. e^-e^+ scattering

Now, consider e^-e^+ scattering. We again have two diagrams contributing. One is similar to $e^-\mu^-$ scattering



We find

$$\mathcal{M}_1 \sim \int \frac{d^4q}{(2\pi)^4} [\bar{u}(3)ig_e\gamma^\mu u(1)] \frac{-i\eta_{\mu\nu}}{q^2} \underbrace{[\bar{v}(2)ig_e\gamma^\nu v(4)]}_{e^+ \text{ line}} \times (2\pi)^4 \delta^{(4)}(p_1 - p_3 - q) (2\pi)^4 \delta^{(4)}(p_2 + q - p_4)$$

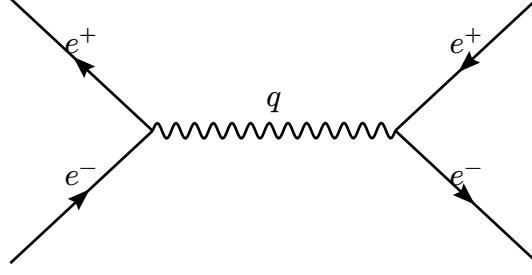
Note, when going “backwards” along an antiparticle we are going forwards in time.⁶⁸

We find

$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3)\gamma^\mu u(1)] [\bar{v}(2)\gamma_\mu v(4)]$$

The second contributing diagram is

⁶⁸The order is always adjoint $\times$ interaction $\times$ spinor when we follow “fermion-flow”.



We find

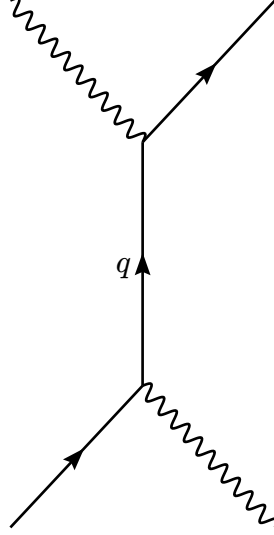
$$\begin{aligned}\mathcal{M}_2 &\sim \int \frac{d^4 q}{(2\pi)^4} [\bar{v}(2) i g_e \gamma^\mu u(1)] \frac{-i \eta_{\mu\nu}}{q^2} [\bar{u}(3) i g_e \gamma^\nu v(4)] \\ &\times (2\pi)^4 \delta^{(4)}(p_1 + p_2 - q) (2\pi)^4 \delta^{(4)}(q - p_3 - p_4) \\ \mathcal{M}_2 &= -\frac{g_e^2}{(p_1 + p_2)^2} [\bar{u}(3) \gamma^\mu v(4)] [\bar{v}(2) \gamma_\mu u(1)]\end{aligned}$$

Here, exchanging the incoming positron e^+ with the outgoing electron e^- we recover the first diagram. We find

$$\begin{aligned}\mathcal{M} &= -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3) \gamma^\mu u(1)] [\bar{v}(2) \gamma_\mu v(4)] \\ &\quad \underbrace{+}_{\text{by rule 8}} \frac{g_e^2}{(p_1 + p_2)^2} [\bar{u}(3) \gamma^\mu v(4)] [\bar{v}(2) \gamma_\mu u(1)]\end{aligned}$$

7.11. Compton scattering

Now, consider $e^- \gamma$ scattering. We again have two diagrams contributing. The first diagram is



We find

$$\mathcal{M}_1 \sim \int \frac{d^4 q}{(2\pi)^4} \epsilon_\mu(2) \overbrace{\left[\overline{u}(4) i g_e \gamma^\mu \frac{i(\not{q} + mc)}{q^2 - m^2 c^2} i g_e \gamma^\nu u(1) \right]}^{\text{sandwich}} \epsilon_\nu(3)^* \\ \times (2\pi)^4 \delta^{(4)}(p_1 - p_3 - q) (2\pi)^4 \delta^{(4)}(p_2 + q - p_4)$$

$\underbrace{\hspace{10em}}_{e^- \text{ line}}$

Here, we contract each ϵ_μ with the index of the vertex where it was “created”. We find

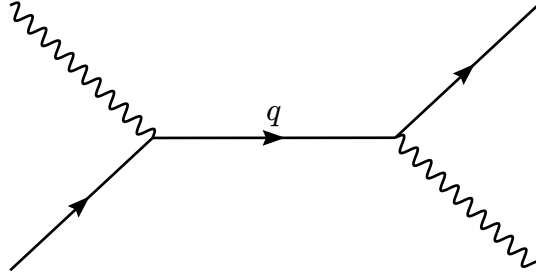
$$\mathcal{M}_1 = \frac{g_e^2}{(p_1 - p_3)^2 - m^2 c^2} [\overline{u}(4) \not{\epsilon}(2) (\not{p}_1 - \not{p}_3 + mc) \not{\epsilon}(3)^* u(1)]$$

where we define

$$\not{\epsilon} \equiv \gamma^\mu a_\mu$$

Note, we will write $\not{\epsilon}^* = \gamma^\mu (\epsilon_\mu^*)$.

The second contributing diagram is



We find

$$\mathcal{M}_2 = \frac{g_e^2}{(p_1 + p_2)^2 - m^2 c^2} [\bar{u}(4) \not{\epsilon}(3)^* (\not{p}_1 + \not{p}_2 + mc) \not{\epsilon}(2) u(1)]$$

7.12. Computing $|\mathcal{M}|^2$

Now, computing $|\mathcal{M}|^2$ requires knowing the spins. We usually do not know the incoming spins s_i and outgoing spins s_f exactly. We instead use the “spin-averaged” amplitude

$$\langle |\mathcal{M}|^2 \rangle \equiv \frac{1}{N_i} \sum_{s_i} \sum_{s_f} |\mathcal{M}(s_i \rightarrow s_f)|^2,$$

where we only average over the initial spins s_i !⁶⁹

Now, consider $e^- \mu^-$ scattering. We found

$$|\mathcal{M}|^2 = \frac{g_e^4}{(p_1 - p_3)^4} [\bar{u}(3) \gamma^\mu u(1)] [\bar{u}(3) \gamma^\nu u(1)]^* [\bar{u}(4) \gamma_\mu u(2)] [\bar{u}(4) \gamma_\nu u(2)]^*$$

Here, “all” terms are of the form

$$G = [\bar{u}(a) \Gamma_1 u(b)] [\bar{u}(a) \Gamma_2 u(b)]^*$$

where $\Gamma_1 = \gamma^\mu$ and $\Gamma_2 = \gamma^\nu$ above. We have

$$[\bar{u}(a) \Gamma_2 u(b)]^* = [u(a)^\dagger \gamma^0 \Gamma_2 u(b)]^\dagger = u(b)^\dagger \Gamma_2^\dagger \gamma^0 u(a) = u(b)^\dagger \underbrace{\gamma^0 \gamma^0}_{\mathbb{1}} \Gamma_2^\dagger \gamma^0 u(a) = \bar{u}(b) \bar{\Gamma}_2 u(a)$$

with $\bar{\Gamma}_2$ being

$$\bar{\Gamma}_2 \equiv \gamma^0 \Gamma_2^\dagger \gamma^0$$

implying

$$G = [\bar{u}(a) \Gamma_1 u(b)] [\bar{u}(b) \bar{\Gamma}_2 u(a)]$$

Now, we sum over s_b

$$\begin{aligned} \sum_{s_b} G &= \bar{u}(a) \Gamma_1 \left[\sum_{s_b} u^{(s_b)}(p_b) \bar{u}^{(s_b)}(p_b) \right] \bar{\Gamma}_2 u(a) \\ &\stackrel{\text{completeness}}{=} \bar{u}(a) \Gamma_1 (\not{p}_b + m_b c) \bar{\Gamma}_2 u(a) \\ &= \bar{u}(a) Q u(a) \end{aligned}$$

⁶⁹When our experimental setup is unaware of the initial spin state and our detector cannot measure outgoing spin.

with Q being

$$Q \equiv \Gamma_1(\not{p}_b + m_b c) \bar{\Gamma}_2$$

Now, we sum over s_a

$$\begin{aligned} \sum_{s_a} \sum_{s_b} G &= \sum_{s_a} \bar{u}^{(s_a)}(p_a) Q u^{(s_a)}(p_a) \\ &= \sum_{s_a} \sum_{\alpha, \beta=1}^4 \bar{u}_{\alpha}^{(s_a)}(p_a) Q_{\alpha\beta} u_{\beta}^{(s_a)}(p_a) \\ &= \sum_{\alpha, \beta=1}^4 Q_{\alpha\beta} \left[\sum_{s_a} u_{\beta}^{(s_a)}(p_a) \bar{u}_{\alpha}^{(s_a)}(p_a) \right] \\ &= \sum_{\alpha, \beta=1}^4 Q_{\alpha\beta} (\not{p}_a + m_a c)_{\beta\alpha} \\ &= \text{tr}[Q(\not{p}_a + m_a c)] \end{aligned}$$

implying

$$\sum_{s_a, s_b} [\bar{u}(a) \Gamma_1 u(b)] [\bar{u}(a) \Gamma_2 u(b)]^* = \text{tr} [\Gamma_1(\not{p}_b + m_b c) \bar{\Gamma}_2(\not{p}_a + m_a c)]$$

which is called Casimir's trick.⁷⁰ When we have done the sums over s_i and s_f we need to remember the N_i^{-1} .

Now, with $e^- \mu^-$ scattering we had $\Gamma_2 = \gamma^\nu$ so

$$\bar{\Gamma}_2 = \gamma^0 \gamma^{\nu\dagger} \gamma^0 = \gamma^\nu$$

We apply Casimir's trick to find

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{\underbrace{4}_{N_i^{-1}}} \frac{g_e^4}{(p_1 - p_3)^4} \underbrace{\text{tr}[\gamma^\mu (\not{p}_1 + m_e c) \gamma^\nu (\not{p}_3 + m_e c)]}_{\text{from } \sum_{s_1, s_3}} \overbrace{\text{tr}[\gamma_\mu (\not{p}_2 + m_\mu c) \gamma_\nu (\not{p}_4 + m_\mu c)]}^{\text{from } \sum_{s_2, s_4}}$$

We recall,

$$\text{tr}(A + B) = \text{tr} A + \text{tr} B, \quad \text{tr} \alpha A = \alpha \text{tr} A, \quad \text{tr} AB = \text{tr} BA$$

⁷⁰Here, we had an expression of the form $\dots \bar{u}(a)[\bullet]u(b)\bar{u}(b)[\bullet]u(a)\dots$. We should think of this expression as being circular i.e. the $\dots$ connect. We can immediately replace $u\bar{u} \rightarrow (\not{p} + mc)$ in this case or $v\bar{v} \rightarrow (\not{p} - mc)$ when we have “mixed” or “antiparticle” cases.

and,

$$\eta_{\mu\nu}\eta^{\mu\nu} = 4$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \Rightarrow \{\not{a}, \not{b}\} = 2a \cdot b$$

With these one can derive the contraction theorems

$$\gamma_\mu \gamma^\mu = 4$$

$$\gamma_\mu \gamma^\nu \gamma^\mu = -2\gamma^\nu \Rightarrow \gamma_\mu \not{a} \gamma^\mu = -2\not{a}$$

$$\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu = 4\eta^{\nu\lambda} \Rightarrow \gamma_\mu \not{a} \not{b} \gamma^\mu = 4a \cdot b$$

$$\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\lambda \gamma^\nu \Rightarrow \gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{a} \not{b} \not{c}$$

and the trace theorems

$$\text{tr odd } \not{a} \gamma^\mu = 0$$

$$\text{tr } 1 = 4$$

$$\text{tr } \gamma^\mu \gamma^\nu = 4\eta^{\mu\nu} \Rightarrow \text{tr } \not{a} \not{b} = 4a \cdot b$$

$$\text{tr } \gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma = 4(\eta^{\mu\nu}\eta^{\lambda\sigma} - \eta^{\mu\lambda}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\lambda})$$

$$\Rightarrow \text{tr } \not{a} \not{b} \not{c} \not{d} = 4[(a \cdot b)(c \cdot d) - (a \cdot c)(b \cdot d) + (a \cdot d)(b \cdot c)]$$

With $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ we also have

$$\text{tr } \gamma^5 = 0$$

$$\text{tr } \gamma^5 \gamma^\mu \gamma^\nu = 0 \Rightarrow \text{tr } \gamma^5 \not{a} \not{b} = 0$$

$$\text{tr } \gamma^5 \gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma = 4i\epsilon^{\mu\nu\lambda\sigma} \Rightarrow \text{tr } \gamma^5 \not{a} \not{b} \not{c} \not{d} = 4i\epsilon^{\mu\nu\lambda\sigma} a_\mu b_\nu c_\lambda d_\sigma$$

Now, consider one of the traces in $e^- \mu$ scattering

$$\text{tr}[\gamma^\mu(\not{p}_1 + m_e c)\gamma^\nu(\not{p}_3 + m_e c)] = \text{tr } \gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3 + m_e c \underbrace{[\text{tr } \gamma^\mu \not{p}_1 \gamma^\nu + \text{tr } \gamma^\mu \gamma^\nu \not{p}_3]}_{0 \text{ by } \# \gamma^\mu \text{ being odd}} + (m_e c)^2 \underbrace{\text{tr } \gamma^\mu \gamma^\nu}_{4\eta^{\mu\nu}}$$

We compute

$$\text{tr } \gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3 = \overbrace{(p_1)_\lambda}^{\text{numbers}} \overbrace{(p_3)_\sigma}^{\text{matrices}} \text{tr } \gamma^\mu \gamma^\lambda \gamma^\nu \gamma^\sigma$$

$$\begin{aligned}
&= (p_1)_\lambda (p_3)_\sigma 4(\eta^{\mu\lambda}\eta^{\nu\sigma} - \eta^{\mu\nu}\eta^{\lambda\sigma} + \eta^{\mu\sigma}\eta^{\lambda\nu}) \\
&= 4[p_1^\mu p_3^\nu - \eta^{\mu\nu}(p_1 \cdot p_3) + p_3^\mu p_1^\nu]
\end{aligned}$$

implying

$$\text{tr}[\gamma^\mu (\not{p}_1 + m_e c) \gamma^\nu (\not{p}_3 + m_e c)] = 4\{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu + \eta^{\mu\nu}[(m_e c)^2 - (p_1 \cdot p_3)]\}$$

We find

$$\begin{aligned}
\langle |\mathcal{M}|^2 \rangle &= \frac{4g_e^4}{(p_1 - p_3)^4} \left\{ p_1^\mu p_3^\nu + p_3^\mu p_1^\nu + \eta^{\mu\nu}[(m_e c)^2 - (p_1 \cdot p_3)] \right\} \left\{ p_{2\mu} p_{4\nu} + p_{4\mu} p_{2\nu} + \eta_{\mu\nu}[(m_\mu c)^2 - (p_2 \cdot p_4)] \right\} \\
&= \frac{8g_e^4}{(p_1 - p_3)^4} \left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(m_\mu c)^2 - (p_2 \cdot p_4)(m_e c)^2 + 2(m_e m_\mu c^2)^2 \right]
\end{aligned}$$

7.13. Γ for $e^- \mu$ scattering

Now, we can compute Γ and σ using $\langle |\mathcal{M}|^2 \rangle$.

We consider $e^- \mu^-$ scattering assuming $m_\mu \gg m_e$ and would like σ in the lab frame with m_μ at rest. We have

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar}{8\pi M c} \right)^2 \langle |\mathcal{M}|^2 \rangle$$

with

$$p_1 = \left(\frac{E}{c}, \mathbf{p}_1 \right); \quad p_2 = (m_\mu c, \mathbf{0}); \quad p_3 = \left(\frac{E}{c}, \mathbf{p}_3 \right); \quad p_4 = (m_\mu c, \mathbf{0})$$

and $|\mathbf{p}_1| = |\mathbf{p}_3| = |\mathbf{p}|$ with $\mathbf{p}_1 \cdot \mathbf{p}_3 = p^2 \cos \theta$. We have

$$(p_1 - p_3)^2 = -(p_1 + p_3)^2 = -4p^2 \sin^2 \frac{\theta}{2}$$

$$(p_1 \cdot p_3) = m_e^2 c^2 + 2p^2 \sin^2 \frac{\theta}{2}$$

$$(p_1 \cdot p_2)(p_3 \cdot p_4) = (p_1 \cdot p_4)(p_2 \cdot p_3) = (m_\mu E)^2$$

$$(p_2 \cdot p_4) = (m_\mu c)^2$$

implying

$$\langle |\mathcal{M}|^2 \rangle = \left(\frac{g_e^2 m_\mu c}{p^2 \sin^2 \theta/2} \right)^2 \left[(m_e c)^2 + p^2 \cos^2 \frac{\theta}{2} \right]$$

and

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha \hbar}{2p^2 \sin^2 \theta/2} \right)^2 \left[(m_e c)^2 + p^2 \cos^2 \frac{\theta}{2} \right]$$

which is called the Mott formula. With $p^2 \ll (m_e c)^2$ we find the Rutherford formula

$$\frac{d\sigma}{d\Omega} = \left(\frac{e^2}{2mv^2 \sin^2 \theta/2} \right)^2$$

which is quite nice.

7.14. Pair annihilation

Now, we consider pair annihilation of the form $e^+ + e^- \rightarrow 2\gamma$. We have two diagrams with⁷¹

$$\mathcal{M}_1 = \frac{g_e^2}{(p_1 - p_3)^2 - m^2 c^2} \bar{v}(2) \not{\epsilon}_4 (\not{p}_1 - \not{p}_3 + mc) \not{\epsilon}_3 u(1)$$

$$\mathcal{M}_2 = \frac{g_e^2}{(p_1 - p_4)^2 - m^2 c^2} \bar{v}(2) \not{\epsilon}_3 (\not{p}_1 - \not{p}_4 + mc) \not{\epsilon}_4 u(1)$$

and $\mathcal{M} = \mathcal{M}_1 + \mathcal{M}_2$.

We will work in the CM-frame and assume the $e^\pm$ are initially $\sim$ at rest. Here, we will be unable to average over s_i since the composite system $e^- e^+$ is in the singlet or triplet configuration. When e^+ and e^- are at rest the photons γ come out back-to-back. We take the z -axis to coincide with the direction of the first photon. We have⁷²

$$p_1 = \begin{pmatrix} mc \\ \mathbf{0} \end{pmatrix}, \quad p_2 = \begin{pmatrix} mc \\ \mathbf{0} \end{pmatrix}, \quad p_3 = \begin{pmatrix} mc \\ 0 \\ 0 \\ mc \end{pmatrix}, \quad p_4 = \begin{pmatrix} mc \\ 0 \\ 0 \\ -mc \end{pmatrix}$$

implying

$$(p_1 - p_3)^2 - m^2 c^2 = (p_1 - p_4)^2 - m^2 c^2 = -2(mc)^2$$

⁷¹One is “twisted”.

⁷²We use $E_i = 2mc^2 \stackrel{!}{=} 2E_\gamma = 2|p|c$

Now, we use

$$\not{p}_1 \not{\epsilon}_3 = -\not{\epsilon}_3 \not{p}_1 + 2 \underbrace{p_1 \cdot \epsilon_3}_{0 \text{ in Coulomb gauge}}$$

$$\not{p}_3 \not{\epsilon}_3 = -\not{\epsilon}_3 \not{p}_3 + 2 \underbrace{p_3 \cdot \epsilon_3}_{0 \text{ by Lorenz gauge}}$$

implying

$$(\not{p}_1 - \not{p}_3 + mc) \not{\epsilon}_3 u(1) = \not{\epsilon}_3 (-\not{p}_1 + \not{p}_3 + mc) u(1)$$

$$\stackrel{\text{by Dirac}}{=} \not{\epsilon}_3 \not{p}_3 u(1)$$

$$(\not{p}_1 - \not{p}_4 + mc) \not{\epsilon}_4 u(1) = \not{\epsilon}_4 \not{p}_4 u(1)$$

We find

$$\mathcal{M} = -\frac{g_e^2}{2(mc)^2} \bar{v}(2) [\not{\epsilon}_4 \not{\epsilon}_3 \not{p}_3 + \not{\epsilon}_3 \not{\epsilon}_4 \not{p}_4] u(1)$$

with

$$\not{p}_3 = mc(\gamma^0 - \gamma^3); \quad \not{p}_4 = mc(\gamma^0 + \gamma^3)$$

implying

$$[\not{\epsilon}_4 \not{\epsilon}_3 \not{p}_3 + \not{\epsilon}_3 \not{\epsilon}_4 \not{p}_4] = mc[(\not{\epsilon}_4 \not{\epsilon}_3 + \not{\epsilon}_3 \not{\epsilon}_4) \gamma^0 - (\not{\epsilon}_4 \not{\epsilon}_3 - \not{\epsilon}_3 \not{\epsilon}_4) \gamma^3]$$

Now, we use

$$\not{\epsilon} = -\epsilon \cdot \gamma = -\begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \boldsymbol{\epsilon} \\ -\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon} & 0 \end{pmatrix}$$

or

$$\not{\epsilon}_3 \not{\epsilon}_4 = -\begin{pmatrix} (\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_3)(\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_4) & 0 \\ 0 & (\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_3)(\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_4) \end{pmatrix}$$

with

$$(\boldsymbol{\sigma} \cdot \boldsymbol{a})(\boldsymbol{\sigma} \cdot \boldsymbol{b}) = \boldsymbol{a} \cdot \boldsymbol{b} + i\boldsymbol{\sigma} \cdot (\boldsymbol{a} \times \boldsymbol{b})$$

implying

$$(\not{\epsilon}_4 \not{\epsilon}_3 + \not{\epsilon}_3 \not{\epsilon}_4) = -2\boldsymbol{\epsilon}_3 \cdot \boldsymbol{\epsilon}_4, \quad (\not{\epsilon}_4 \not{\epsilon}_3 - \not{\epsilon}_3 \not{\epsilon}_4) = 2i(\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4) \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$$

We find

$$\mathcal{M} = \frac{g_e^2}{mc} \bar{v}(2) \left[(\epsilon_3 \cdot \epsilon_4) \gamma^0 + i(\epsilon_3 \times \epsilon_4) \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix} \gamma^3 \right] u(1)$$

Now, we consider the singlet configuration of e^-e^+ or symbolically

$$\mathcal{M}_{\text{singlet}} = \frac{1}{\sqrt{2}} (\mathcal{M}_{\uparrow\downarrow} - \mathcal{M}_{\downarrow\uparrow})$$

where $\mathcal{M}_{\uparrow\downarrow}$ is found by using $u^{(1)}$ (spin up e^-) and $v^{(2)}$ (spin down e^+) given by

$$u(1) = \sqrt{2mc} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \bar{v}(2) = \sqrt{2mc} (0, 0, 1, 0)$$

We find

$$\bar{v}(2) \gamma^0 u(1) = 0, \quad \bar{v}(2) \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix} \gamma^3 u(1) = -2mc \hat{z}$$

implying

$$\mathcal{M}_{\uparrow\downarrow} = -2ig_e^2 (\epsilon_3 \times \epsilon_4)_z$$

Likewise, we find

$$\mathcal{M}_{\downarrow\uparrow} = -\mathcal{M}_{\uparrow\downarrow}$$

implying

$$\mathcal{M}_{\text{singlet}} = -2\sqrt{2}ig_e^2 (\epsilon_3 \times \epsilon_4)_z$$

Note, the triplet configuration would have

$$\mathcal{M}_{\text{triplet}} = \frac{1}{\sqrt{2}} (\mathcal{M}_{\uparrow\downarrow} + \mathcal{M}_{\downarrow\uparrow}) = 0$$

implying the decay is forbidden in this case!

Now, we need the corresponding ϵ . Recall the “spin-up” ($m_s = +1$) and “spin-down” ($m_s = -1$) polarisations

$$\epsilon_+ = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}; \quad \epsilon_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

We know the value of L_z should vanish, so the photon spins should be antiparallel $\uparrow\downarrow$ or $\downarrow\uparrow$. When $\uparrow\downarrow$ we have⁷³

$$\epsilon_3 = \epsilon_+, \quad \epsilon_4 = \epsilon_-$$

implying

$$\epsilon_3 \times \epsilon_4 = i\hat{k}$$

When $\downarrow\uparrow$ we have

$$\epsilon_3 \times \epsilon_4 = -i\hat{k}$$

We need the antisymmetric combination⁷⁴

$$\mathcal{M}_{\text{singlet}} = \frac{1}{\sqrt{2}}(\mathcal{M}_{\uparrow\downarrow} - \mathcal{M}_{\downarrow\uparrow}) = -4g_e^2$$

since $\mathcal{M}_{\text{triplet}} = 0$.

Now, we have in the CM-frame

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} \left(\frac{\hbar c}{8\pi(E_1 + E_2)} \right)^2 \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\mathcal{M}|^2$$

with

$$E_1 = E_2 = mc^2, \quad |\mathbf{p}_f| = mc, \quad |\mathbf{p}_i| = \frac{mv}{2}$$

where v is the relative velocity between e^- and e^+ . We find

$$\frac{d\sigma}{d\Omega} = \frac{1}{cv} \left(\frac{\hbar\alpha}{m} \right)^2 \Rightarrow \sigma = \frac{4\pi}{cv} \left(\frac{\hbar\alpha}{m} \right)^2$$

Now, we can compute $\tau_{e^-e^+}$. We can write

$$\frac{d\sigma}{d\Omega} = \frac{1}{\mathcal{L}} \frac{dN}{d\Omega}$$

or $N = \mathcal{L}\sigma$ with

$$\mathcal{L} = \rho v$$

where ρ is the number density of incident particles. The e^- density of a single “atom” is $|\psi(0)|^2$ and $N = \Gamma$ implying

⁷³Here, the photon 4 has $h = +1$ along $-z$ i.e. $\epsilon_+(+z) = \epsilon_-(-z)$

⁷⁴Where $\uparrow$ and $\downarrow$ refer to photon polarisation.

$$\Gamma = v\sigma|\psi(0)|^2 = \frac{4\pi}{c} \left(\frac{\hbar\alpha}{m} \right)^2 |\psi(0)|^2$$

Now, the ground state has

$$|\psi(0)|^2 = \frac{1}{\pi} \left(\frac{\alpha mc}{2\hbar} \right)^2$$

implying

$$\tau = \Gamma^{-1} = \frac{2\hbar}{\alpha^5 mc^2} \sim 10^{-10} \text{ s}$$